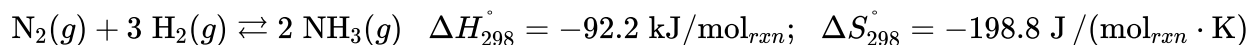


Chapter 10 Part 2: Le Chatelier's Principle FRQ

1. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to NH_3 .



(a) The reaction of $\text{N}_2(g)$ and $\text{H}_2(g)$ to form $\text{NH}_3(g)$ is represented above. The reaction has been studied in order to maximize the yield of $\text{NH}_3(g)$.

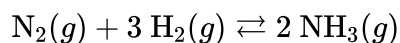
- (i) Calculate the value of ΔG° , in kJ/mol_{rxn} , at 298 K.
- (ii) Calculate the value of the equilibrium constant, K_p , at 298 K.

Bond	Average Bond Enthalpy (kJ/mol)
N – H	391
H – H	436
N $\equiv$ N	?

(b) Based on the average bond enthalpies in the table above and the value of ΔH_{298}° , calculate the value of the N $\equiv$ N bond enthalpy.

(c) Although the reaction is thermodynamically favorable at 298 K, it does not occur to an observable extent at this temperature. Explain how the N $\equiv$ N bond enthalpy affects the rate of the reaction.

(d) The reaction is usually carried out at a very high temperature in the presence of a catalyst. Explain why the reaction is thermodynamically unfavorable at very high temperatures. Justify your answer in terms of ΔH° , ΔS° , and their effect on ΔG° .



(e) Although the forward reaction represented above is thermodynamically unfavorable at very high temperatures, it can be used to produce NH_3 if the NH_3 is continually removed from the reaction vessel. Explain in terms of Q and K_p why this method works.

The boiling points of H_2 and NH_3 are given in the table below.

Substance	Boiling Point
H_2	20 K
NH_3	240 K

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(f) A student claims that $\text{H}_2(g)$ condenses at a much lower temperature than $\text{NH}_3(g)$ because H_2 has stronger intermolecular forces than NH_3 does. There are two errors in the student's reasoning. Identify the student's errors and explain your answer in terms of the specific types of forces involved.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **all** the following criteria.

- ☐ The response provides a correct calculated value:
 $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
 $\Delta G^\circ = -82.2 \text{ kJ/mol}_{rxn} - (298 \text{ K})(-0.1988 \text{ kJ}/(\text{K} \cdot \text{mol}_{rxn}))$
 $\Delta G^\circ = -33.0 \text{ kJ/mol}_{rxn}$
- ☐ The response provides the correct calculated value consistent with the calculated value for ΔG° :
 $\Delta G^\circ = -RT \ln K_p$
 $\ln K_p = \frac{-\Delta G^\circ}{RT} = \frac{-(-33,000 \text{ J/mol}_{rxn})}{(8.314 \frac{\text{J}}{\text{K} \cdot \text{mol}_{rxn}})(298 \text{ K})} = 13.3$
 $K_p = e^{13.3} = 6.09 \times 10^5$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **all** the following criteria.

- ☐ The response identifies endothermic and exothermic bond enthalpies:
 $\Delta H^\circ_{rxn} = \sum(\Delta H^\circ_{bonds\ broken}) - \sum(\Delta H^\circ_{bonds\ formed})$
 OR
 · Assignment of positive and negative signs to bond enthalpies (see the following sample response)
- ☐ The response provides the correct calculated value:

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Bonds Broken	Bonds Formed
$\text{N} \equiv \text{N} \quad x$	$6 \text{ N} - \text{H} \quad 6 (- 391)$
$3 \text{ H} - \text{H} \quad 3(436)$	
$x + 1308 \text{ kJ/mol} + (- 2346 \text{ kJ/mol}) = - 92.2 \text{ kJ/mol}_{rxn}$ $x = 946 \text{ kJ/mol}$	

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation equivalent to the following.

· The large bond energy of the $\text{N} \equiv \text{N}$ bond contributes to a high activation energy for the reaction, which causes the reaction to proceed slowly.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **all** the following criteria.

- ☐ The response provides a correct explanation of thermodynamic favorability.
 - For any thermodynamically unfavorable process, the sign of ΔG° will be positive.
- ☐ The response provides a valid justification based on the relative magnitudes of ΔH° and ΔS° .
 - At high temperature, the $-T\Delta S^\circ$ term becomes a larger positive value and eventually becomes larger in magnitude than the ΔH° term, causing the ΔG° value to become positive

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$Q = \frac{(P_{\text{NH}_3})^2}{(P_{\text{N}_2})(P_{\text{H}_2})^3}$$

· Removing NH_3 from the reaction vessel reduces the value of Q . The reaction consumes more reactants to generate more NH_3 because $Q < K$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

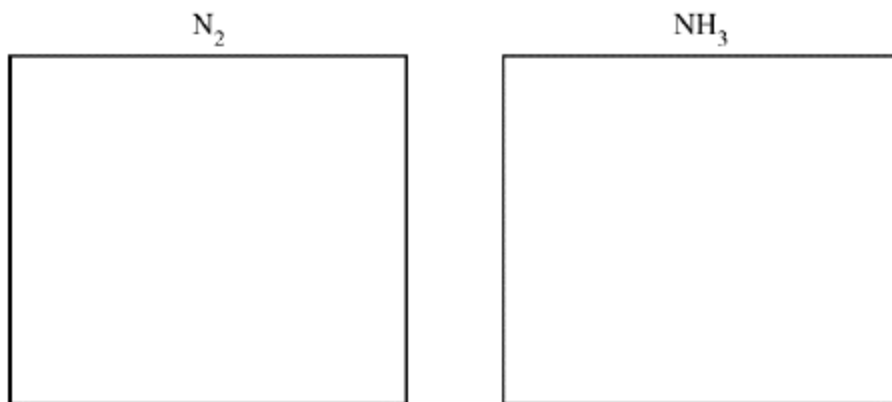
The response meets **all** the following criteria.

- ☐ The response provides a correct explanation of the boiling point and strength of intermolecular forces.
 - The student has reversed the relationship between a substance's boiling point (condensation temperature) and the strength of its intermolecular forces. A lower boiling point indicates that the intermolecular forces between particles are weaker.
- ☐ The response provides a correct explanation based on the intermolecular forces in each substance.
 - The student has misjudged the relative strength of intermolecular forces of the two substances. Molecules of H_2 experience only London dispersion forces, while molecules of NH_3 experience dipole-dipole forces and hydrogen bonding in addition to London dispersion forces. Therefore, H_2 has weaker intermolecular forces than NH_3 does.

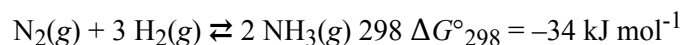
Chapter 10 Part 2: Le Chatelier's Principle FRQ

2. Answer the following questions about nitrogen, hydrogen, and ammonia.

- a. In the boxes below, draw the complete Lewis electron-dot diagrams for N_2 and NH_3 .



- b. Calculate the standard free-energy change, ΔG° , that occurs when 12.0 g of $\text{H}_2(\text{g})$ reacts with excess $\text{N}_2(\text{g})$ at 298 K according to the reaction represented below.

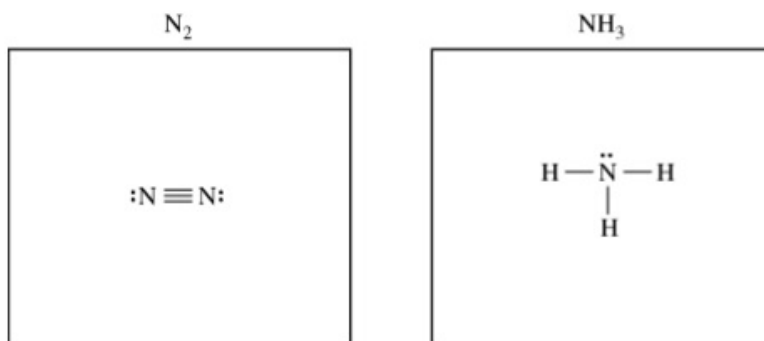


- c. Given that ΔH°_{298} for the reaction is $-92.2 \text{ kJ mol}^{-1}$, which is larger, the total bond dissociation energy of the reactants or the total bond dissociation energy of the products? Explain.
- d. The value of the standard entropy change, ΔS°_{298} , for the reaction is $-199 \text{ J mol}^{-1} \text{K}^{-1}$. Explain why the value of ΔS°_{298} is negative.
- e. Assume that ΔH° and ΔS° for the reaction are independent of temperature.
- Explain why there is a temperature above 298 K at which the algebraic sign of the value of ΔG° changes.
 - Theoretically, the best yields of ammonia should be achieved at low temperatures and high pressures.

Part A

2 points are earned for the correct Lewis electron-dot diagrams (1 point each).

Chapter 10 Part 2: Le Chatelier's Principle FRQ



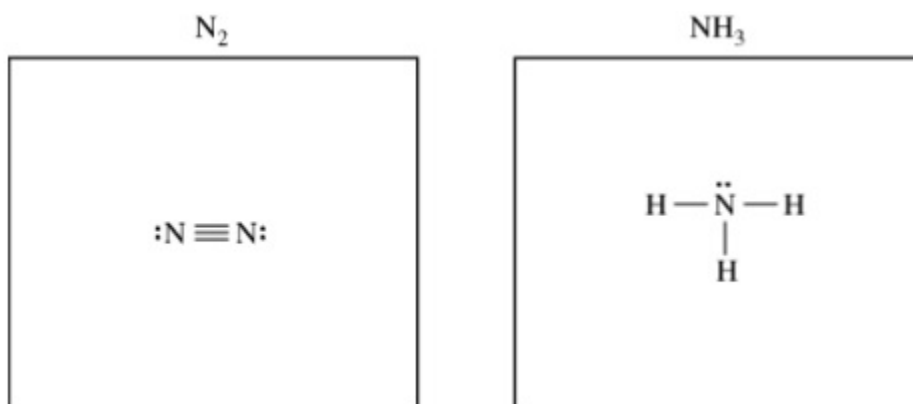
The correct structures are shown in the boxes above.



0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for the correct Lewis electron-dot diagrams (1 point each).



The correct structures are shown in the boxes above.

Part B

1 point is earned for the correct stoichiometry.

1 point is earned for the correct answer.

$$12.0 \text{ g H}_2 \times \frac{1 \text{ mol H}_2}{2.0 \text{ g H}_2} \times \frac{1 \text{ mol reaction}}{3 \text{ mol H}_2} \times \frac{-34 \text{ kJ}}{1 \text{ mol reaction}} = -68 \text{ kJ}$$



0	1	2
---	---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The student response earns two of the following points:

1 point is earned for the correct stoichiometry.

1 point is earned for the correct answer.

$$12.0 \text{ g H}_2 \times \frac{1 \text{ mol H}_2}{2.0 \text{ g H}_2} \times \frac{1 \text{ mol reaction}}{3 \text{ mol H}_2} \times \frac{-34 \text{ kJ}}{1 \text{ mol reaction}} = -68 \text{ kJ}$$

Part C

1 point is earned for the correct answer with the correct equation and explanation.

$\Delta H^\circ_{298} = \Sigma (\text{bond energy of the reactants}) - \Sigma (\text{bond energy of the products})$ Based on the equation above, for ΔH°_{298} to be negative, the total bond energy of the products must be larger than the total bond energy of the reactants.

OR

More energy is released as product bonds are formed than is absorbed as reactant bonds are broken.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with the correct equation and explanation.

$\Delta H^\circ_{298} = \Sigma (\text{bond energy of the reactants}) - \Sigma (\text{bond energy of the products})$ Based on the equation above, for ΔH°_{298} to be negative, the total bond energy of the products must be larger than the total bond energy of the reactants.

OR

More energy is released as product bonds are formed than is absorbed as reactant bonds are broken.

Part D

1 point is earned for the correct explanation.

All of the reactants and products in the reaction are in the gas phase, so the sign of the entropy change will depend on the number of moles of particles in the reactants and products. There are more moles of reactants (four) compared with moles of products (two), so there is a greater number of microstates in the reactants than in the products. Therefore the entropy decreases as the reaction proceeds (fewer possible microstates), and the sign of the entropy change is negative.



0	1
---	---

The student response earns one of the following points:

Chapter 10 Part 2: Le Chatelier's Principle FRQ

1 point is earned for the correct explanation.

All of the reactants and products in the reaction are in the gas phase, so the sign of the entropy change will depend on the number of moles of particles in the reactants and products. There are more moles of reactants (four) compared with moles of products (two), so there is a greater number of microstates in the reactants than in the products. Therefore the entropy decreases as the reaction proceeds (fewer possible microstates), and the sign of the entropy change is negative.

Part E

- 1 point is earned for the correct explanation.

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ As the temperature increases $|T\Delta S^\circ|$ will at some point exceed $|\Delta H^\circ|$. Because both ΔH° and ΔS° are negative, the sign of ΔG° will then change from negative to positive.

- 1 point is earned for explaining increased yield at low temperatures.

Low temperatures: The reaction is exothermic. By Le Chatelier's principle, decreasing the temperature drives the reaction to the right to produce more heat energy, and thus more ammonia is produced.

1 point is earned for explaining increased yield at high pressure.

High pressures: For this reaction, higher pressure is achieved by decreasing the volume of the container. As pressure increases, the reaction equilibrium shifts in the direction that reduces the total number of particles (by Le Chatelier's principle). In this case, the product has fewer moles of particles than the reactants; thus product would be favored. Higher pressure therefore results in an increase in the amount of ammonia.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

- 1 point is earned for the correct explanation.

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ As the temperature increases $|T\Delta S^\circ|$ will at some point exceed $|\Delta H^\circ|$. Because both ΔH° and ΔS° are negative, the sign of ΔG° will then change from negative to positive.

- 1 point is earned for explaining increased yield at low temperatures.

Low temperatures: The reaction is exothermic. By Le Chatelier's principle, decreasing the temperature drives the reaction to the right to produce more heat energy, and thus more ammonia is produced.

1 point is earned for explaining increased yield at high pressure.

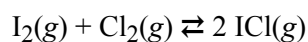
High pressures: For this reaction, higher pressure is achieved by decreasing the volume of the container. As pressure increases, the reaction equilibrium shifts in the direction that reduces the total number of particles (by Le Chatelier's principle). In this case, the product has fewer moles of particles than the reactants; thus product would be favored. Higher pressure therefore results in an increase in the amount of ammonia.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

3. Answer the following questions about the thermodynamics of the reactions represented below.



- Is reaction X, represented above, spontaneous under standard conditions? Justify your answer with a calculation.
- Calculate the value of the equilibrium constant, K_{eq} , for reaction X at 25°C.
- What effect will an increase in temperature have on the equilibrium constant for reaction X? Explain your answer.
- Explain why the standard entropy change is greater for reaction Y than for reaction X.
- Above what temperature will the value of the equilibrium constant for reaction Y be greater than 1.0? Justify your answer with calculations.
- For the vaporization of solid iodine, $\text{I}_2(s) \rightarrow \text{I}_2(g)$, the value of ΔH_{298}° is 62 kJ mol^{-1} . Using this information, calculate the value of ΔH_{298}° for the reaction represented below.



Part A

1 point is earned for the correct value of ΔG° .

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= (18 \text{ kJ mol}^{-1}) - (298 \text{ K})(0.078 \text{ kJ mol}^{-1} \text{ K}^{-1}) = -5 \text{ kJ mol}^{-1}$$

1 point is earned for a correct justification of spontaneity.

Reaction is spontaneous because $\Delta G^\circ < 0$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct value of ΔG° .

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= (18 \text{ kJ mol}^{-1}) - (298 \text{ K})(0.078 \text{ kJ mol}^{-1} \text{ K}^{-1}) = -5 \text{ kJ mol}^{-1}$$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

1 point is earned for a correct justification of spontaneity.

Reaction is spontaneous because $\Delta G^\circ < 0$.

Part B

1 point is earned for the correct answer.

$$\Delta G^\circ = -RT \ln K_{eq} \Rightarrow \ln K_{eq} = -\frac{\Delta G^\circ}{RT}$$

$$\ln K_{eq} = -\frac{(-5 \text{ kJ mol}^{-1})(10^3 \text{ J kJ}^{-1})}{(8.31 \text{ J mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 2.019$$

$$K_{eq} = e^{2.019} = (7.5314) = 8$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer.

$$\Delta G^\circ = -RT \ln K_{eq} \Rightarrow \ln K_{eq} = -\frac{\Delta G^\circ}{RT}$$

$$\ln K_{eq} = -\frac{(-5 \text{ kJ mol}^{-1})(10^3 \text{ J kJ}^{-1})}{(8.31 \text{ J mol}^{-1} \text{ K}^{-1})(298 \text{ K})} = 2.019$$

$$K_{eq} = e^{2.019} = (7.5314) = 8$$

Part C

1 point is earned for the correct choice with a correct explanation.

$$\Delta G^\circ = -RT \ln K_{eq} = \Delta H^\circ - T\Delta S^\circ \Rightarrow \ln K_{eq} = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Since ΔH° is positive, an increase in T will cause $-\Delta H^\circ/RT$ to become a smaller negative number, therefore K_{eq} will increase.

OR

The reaction is endothermic ($\Delta H = +18 \text{ kJ mol}^{-1}$); an increase in temperature shifts the reaction to favor more products relative to the reactants, resulting in an increase in the value of K_{eq} .



0	1
---	---

The student response earns one of the following points:

Chapter 10 Part 2: Le Chatelier's Principle FRQ

1 point is earned for the correct choice with a correct explanation.

$$\Delta G^\circ = -RT \ln K_{eq} = \Delta H^\circ - T\Delta S^\circ \Rightarrow \ln K_{eq} = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

Since ΔH° is positive, an increase in T will cause $-\Delta H^\circ/RT$ to become a smaller negative number, therefore K_{eq} will increase.

OR

The reaction is endothermic ($\Delta H = +18 \text{ kJ mol}^{-1}$); an increase in temperature shifts the reaction to favor more products relative to the reactants, resulting in an increase in the value of K_{eq} .

Part D

1 point is earned for a correct explanation.

Both reaction X and reaction Y have solid iodine as a reactant, but the second reactant in reaction X is chlorine gas whereas the second reactant in reaction Y is liquid bromine. Liquids have lower entropies than gases, thus in reaction Y the reactants are more ordered (and have lower entropies) than in reaction X . The products of both reaction X and reaction Y have about the same disorder, so the change in entropy from reactants to products is greater in reaction Y than in reaction X .



0	1
---	---

The student response earns one of the following points:

1 point is earned for a correct explanation.

Both reaction X and reaction Y have solid iodine as a reactant, but the second reactant in reaction X is chlorine gas whereas the second reactant in reaction Y is liquid bromine. Liquids have lower entropies than gases, thus in reaction Y the reactants are more ordered (and have lower entropies) than in reaction X . The products of both reaction X and reaction Y have about the same disorder, so the change in entropy from reactants to products is greater in reaction Y than in reaction X .

Part E

1 point is earned for $\Delta G^\circ = 0$.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$K_{eq} = 1 \text{ when } \Delta G^\circ = 0 \Rightarrow T\Delta S^\circ = \Delta H^\circ \Rightarrow$$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{41 \text{ kJ mol}^{-1}}{0.124 \text{ kJ mol}^{-1}\text{K}^{-1}} = 330 \text{ K}$$

1 point is earned for the correct temperature.

So when $T > 330 \text{ K}$, $\Delta G^\circ < 0 \text{ kJ mol}^{-1} \Rightarrow K_{eq} > 1.0$

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for $\Delta G^\circ = 0$.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$K_{eq} = 1 \text{ when } \Delta G^\circ = 0 \Rightarrow T\Delta S^\circ = \Delta H^\circ \Rightarrow$$

$$T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{41 \text{ kJ mol}^{-1}}{0.124 \text{ kJ mol}^{-1}\text{K}^{-1}} = 330 \text{ K}$$

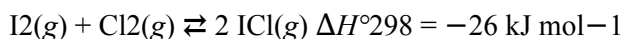
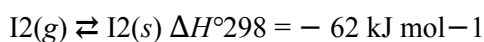
1 point is earned for the correct temperature.

So when $T > 330 \text{ K}$, $\Delta G^\circ < 0 \text{ kJ mol}^{-1} \Rightarrow K_{eq} > 1.0$

Part F

1 point is earned for ΔH°_{298} of either the first or second equation.

1 point is earned for the correct sum of the ΔH°_{298} values.

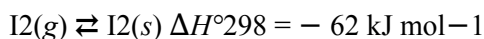


0	1	2
---	---	---

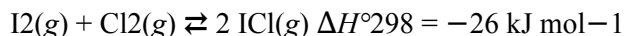
The student response earns two of the following points:

1 point is earned for ΔH°_{298} of either the first or second equation.

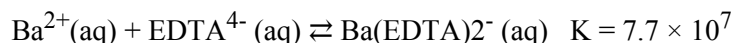
1 point is earned for the correct sum of the ΔH°_{298} values.



Chapter 10 Part 2: Le Chatelier's Principle FRQ



4.



The polyatomic ion $\text{C}_{10}\text{H}_{12}\text{N}_2\text{O}_8^{4-}$ is commonly abbreviated as EDTA^{4-} . The ion can form complexes with metal ions in aqueous solutions. A complex of EDTA^{4-} with Ba^{2+} ion forms according to the equation above. A 50.0 mL volume of a solution that has an $\text{EDTA}^{4-}(\text{aq})$ concentration of 0.30 M is mixed with 50.0 mL of 0.20 M $\text{Ba}(\text{NO}_3)_2$ to produce 100.0 mL of solution.

- Considering the value of K for the reaction, determine the concentration of $\text{Ba}(\text{EDTA})^{2-}(\text{aq})$ in the 100.0 mL of solution. Justify your answer.
- The solution is diluted with distilled water to a total volume of 1.00 L. After equilibrium has been reestablished, is the number of moles $\text{Ba}^{2+}(\text{aq})$ of present in the solution greater than, less than, or equal to the number of moles of $\text{Ba}^{2+}(\text{aq})$ present in the original solution before it was diluted? Justify your answer.

Part A

2 point(s) maximum

1 point is earned for indicating that the equilibrium concentration of $\text{Ba}(\text{EDTA})^{2-}(\text{aq})$ is the same as the original concentration of Ba^{2+} when the solutions are mixed.

Based on the K value, the reaction goes essentially to completion. $\text{Ba}^{2+}(\text{aq})$ is the limiting reactant.

The concentration of Ba^{2+} when the solutions are first mixed but before any reaction takes place is $0.20 \text{ M}/2 = 0.10 \text{ M}$.

1 point is earned for the concentration with appropriate calculations.

Thus the equilibrium concentration of $\text{Ba}(\text{EDTA})^{2-}(\text{aq})$ is 0.10 M.



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for indicating that the equilibrium concentration of $\text{Ba}(\text{EDTA})^{2-}(\text{aq})$ is the same as the original concentration of Ba^{2+} when the solutions are mixed.

Based on the K value, the reaction goes essentially to completion. $\text{Ba}^{2+}(\text{aq})$ is the limiting reactant.

The concentration of Ba^{2+} when the solutions are first mixed but before any reaction takes place is $0.20 \text{ M}/2 =$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0.10 M.

1 point is earned for the concentration with appropriate calculations.

Thus the equilibrium concentration of $\text{Ba}(\text{EDTA})_2^{2-}(\text{aq})$ is 0.10 M.

Part B**2 point(s) maximum**

1 point is earned for stating that the number of moles of $\text{Ba}^{2+}(\text{aq})$ will increase.

1 point is earned for a valid justification.

The number of moles of $\text{Ba}^{2+}(\text{aq})$ increases because the percent dissociation of $\text{Ba}(\text{EDTA})_2^{2-}(\text{aq})$ increases as the solution is diluted.

OR

A mathematical justification such as the following:

The dilution from 100.0 mL to 1.00 L reduces the concentrations of all species to one tenth of their original values.

Immediately after the dilution, the reaction quotient, Q , can be determined as shown below

$$Q = \frac{\frac{1}{10}[\text{Ba}(\text{EDTA})_2^{2-}]}{\frac{1}{10}[\text{Ba}^{2+}] \times \frac{1}{10}[\text{EDTA}^{4-}]} = 10K$$

Because $Q > K$, the net reaction will produce more reactants to move toward equilibrium, so the number of moles of $\text{Ba}^{2+}(\text{aq})$ will be greater than the number in the original solution.



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

2 point(s) maximum

1 point is earned for stating that the number of moles of $\text{Ba}^{2+}(\text{aq})$ will increase.

1 point is earned for a valid justification.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The number of moles of $\text{Ba}^{2+}(aq)$ increases because the percent dissociation of $\text{Ba}(\text{EDTA})^{2-}(aq)$ increases as the solution is diluted.

OR

A mathematical justification such as the following:

The dilution from 100.0 mL to 1.00 L reduces the concentrations of all species to one tenth of their original values.

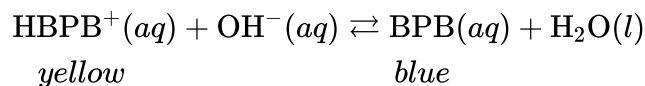
Immediately after the dilution, the reaction quotient, Q , can be determined as shown below

$$Q = \frac{\frac{1}{10}[\text{Ba}(\text{EDTA})^{2-}]}{\frac{1}{10}[\text{Ba}^{2+}] \times \frac{1}{10}[\text{EDTA}^{4-}]} = 10K$$

Because $Q > K$, the net reaction will produce more reactants to move toward equilibrium, so the number of moles of $\text{Ba}^{2+}(aq)$ will be greater than the number in the original solution.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

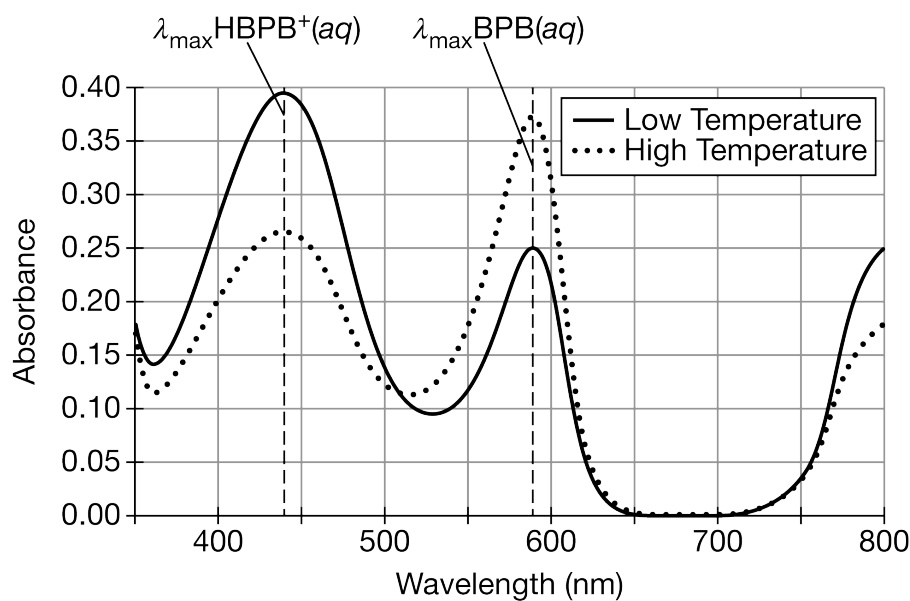
5. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



HBPB^+ , the acidified form of bromophenol blue, BPB, is yellow and has a strong absorption peak at a wavelength of 440 nm. HBPB^+ reacts with hydroxide ions according to the equation above, producing water and BPB, which is blue and has a strong absorption peak at a wavelength of 590 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

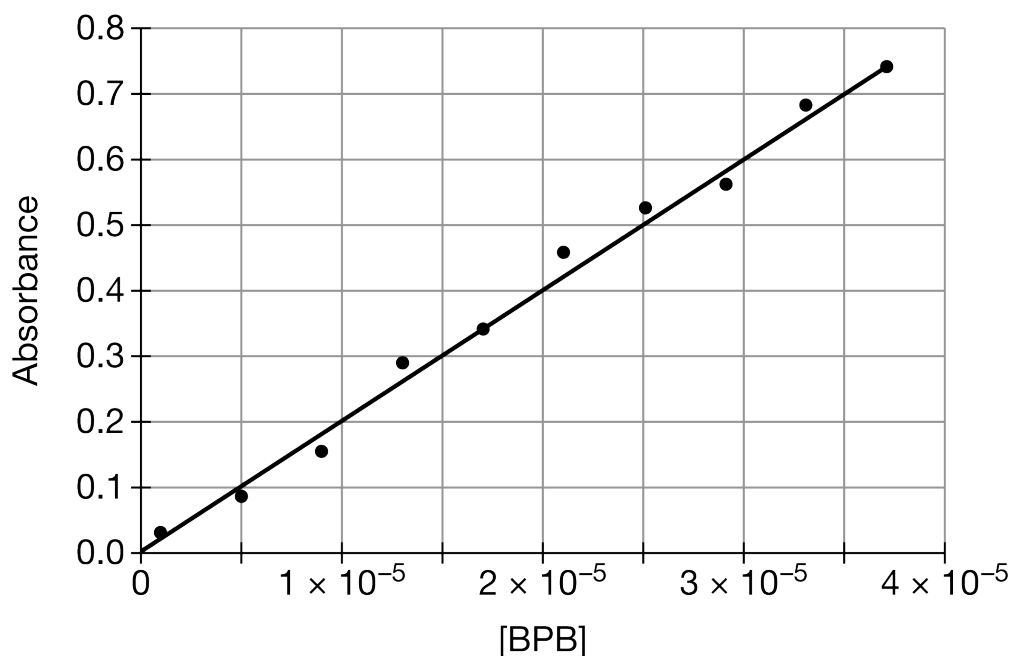
The absorption spectra of an equilibrium mixture of $\text{HBPB}^+(aq)$ and $\text{BPB}(aq)$ collected at two different temperatures, are shown in the diagram below.



- (b) At which wavelength, 350 nm or 440 nm, are photons with the higher energy absorbed? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line shown in the graph, what is the approximate absorbance for a solution in which $[\text{BPB}] = 2.0 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of BPB of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above. If water remains in the cuvette prior to being filled with the solution, how will the estimated molarity compare to the actual $[\text{BPB}]$ in the solution? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 0.10 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (hr)	Absorbance
0	0.80
0.50	0.56
1.00	0.40
1.50	0.28
2.00	0.20
2.50	0.15
3.00	0.10

(f) Determine the half-life of the reaction. Include units with your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 0.40 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OH}^-]$ was 0.10 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OH}^-]$ is changed from 0.10 M to 0.40 M .

(i) When $[\text{OH}^-]$ is changed from 0.10 M to 0.40 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the 0.100 M solution of $\text{NaOH}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 40.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a 1000.0 mL volumetric flask that is half-filled with distilled water.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

(j) The chemical quantities used in the procedure do NOT produce $0.100\text{ M NaOH}(aq)$. Describe specifically how they should be changed in order to produce $0.100\text{ M NaOH}(aq)$.

(k) Another student is preparing a 0.100 M NaOH solution using the correct chemical quantities, but notices that some solid remains at the bottom of the volumetric flask. Describe an action that the student should take to ensure that the remaining solid dissolves.

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in $1000.\text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y?

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

Solvent	X	Y
Specific Heat Capacity (J/(g · °C))	2.53	2.91

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{BPB}]}{[\text{HBPB}^+][\text{OH}^-]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides an explanation equivalent to the following.

· 350 nm. Photons with a wavelength of 350 nm have a higher frequency than those with a wavelength of 440 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the higher frequency (shorter wavelength) photons have the higher energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

· Endothermic. The diagram shows that the absorbance, and therefore the concentration, of BPB is greater at high temperature. The increased concentration of products at elevated temperature is consistent with an endothermic reaction, according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· 0.40

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly describes the effect of the error.

· The estimated molarity of BPB would be too low. The residual water in the cuvette would slightly dilute the solution. Because absorbance and concentration are directly proportional, the diluted solution would give a lower absorbance reading than the undiluted solution and thus lead to an estimated concentration of BPB that is too low.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- ☐ The response indicates a value of 1.00 for the half-life
- ☐ The response has units of “hr” or “hour”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OH^- increases, the collisions between reactant particles occur more frequently per unit of time. The greater frequency of collisions leads to an increased rate of product formation and correspondingly a shorter half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· The magnitude of ΔH_{rxn}° will remain the same. ΔH_{rxn}° is an intensive property, depending on both the enthalpy change and the number of moles reacted. Regardless of the number of moles that react, the ratio of the enthalpy change per mole will always be the same.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

OR

· The magnitude of ΔH_{rxn}° will remain the same. Although the $[\text{OH}^-]$ is increased, the number of moles of each reactant that is consumed is the same, as OH^- is not the limiting reactant. Hence, the magnitude of the enthalpy change per mole would be the same.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid correction to the chemical quantities.

· To make 0.100 M NaOH(aq), the student needs to dissolve 4.00 g of NaOH (0.100 mol) in 1.00 L of solution.

OR

· To make a 0.100 M solution, the student needs to dissolve 0.400 g of NaOH (0.0100 mol) in 100.0 mL of solution.

OR

· This procedure makes a 1.00 M solution, so the student needs to dilute it by a factor of 10.

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid experimental procedure.

· Gently swirl or stir the solution to dissolve the remaining solid in the flask.

OR

· Stopper the flask and invert several times until the solid dissolves.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0

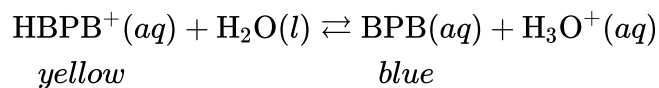
1

The response provides both the correct choice and a valid justification equivalent to the following.

· The temperature change would be greater in solvent X than in solvent Y. Because X has the smaller heat capacity, the same energy transferred to an equal mass of it will result in a greater temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

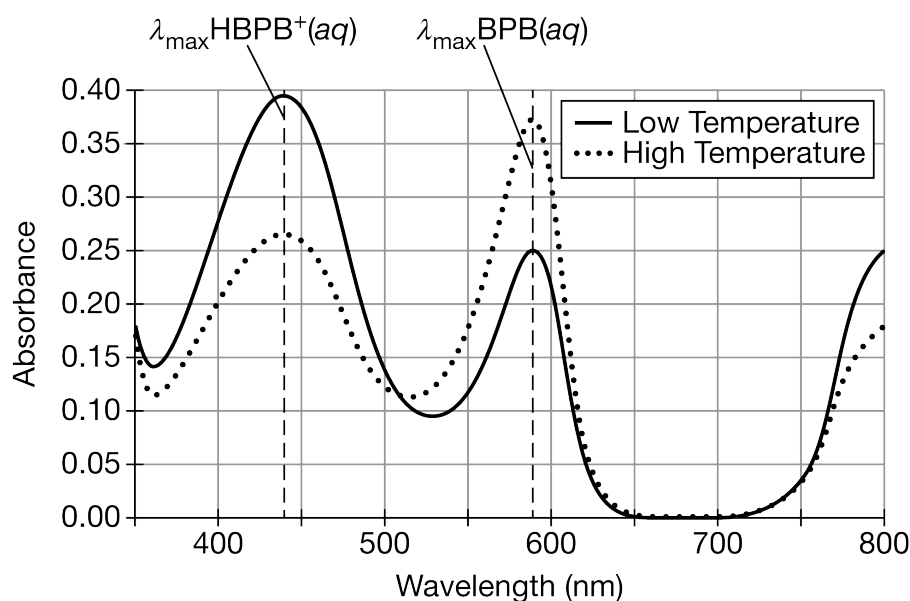
6. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



HBPB^+ , the acidified form of bromophenol blue, BPB, is yellow and has a strong absorption peak at a wavelength of 440 nm. HBPB^+ reacts with water according to the equation above, producing H_3O^+ ions and BPB, which is blue and has a strong absorption peak at a wavelength of 590 nm.

(a) Write the K_{eq} expression for the equilibrium reaction above.

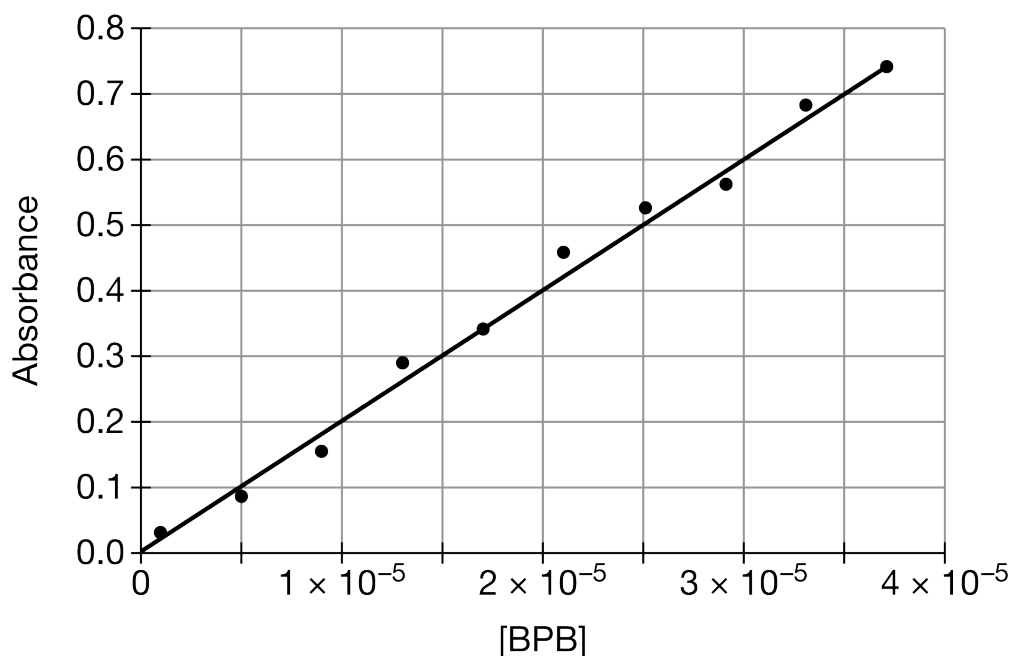
The absorption spectra of an equilibrium mixture of $\text{HBPB}^+(aq)$ and $\text{BPB}(aq)$, collected at two different temperatures, are shown the following diagram.



- (b) Which peak, the one at 590 nm or the one at 800 nm, results from the absorption of photons with the lower energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line shown in the graph, what is the approximate absorbance of a solution in which $[\text{BPB}] = 1.5 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of BPB of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above, which was recorded at 590 nm. If the spectrophotometer is set to a wavelength different from 590 nm during the measurement, what would be the effect on the estimated molarity of BPB? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 1.0 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (s)	Absorbance
0	0.60
120	0.42
240	0.30
360	0.21
480	0.15
600	0.11
720	0.08

(f) Determine the half-life of the reaction. Include units with your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 0.50 M . The observed half-life of the reaction is longer than for the first trial when $[\text{OH}^-]$ was 1.0 M .

(h) Explain, in terms of particle collisions, why the half-life of the reaction increases when $[\text{OH}^-]$ is changed from 1.0 M to 0.50 M .

(i) Is the activation energy for the reaction when $[\text{OH}^-]$ is 0.50 M greater than, the same as, or less than the activation energy when $[\text{OH}^-]$ is 1.0 M ? Justify your answer.

In order to prepare the 0.500 M solution of $\text{NaOH}(aq)$, the student follows a procedure with the following steps.

Step 1: Measure out 20.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a clean, dry 1000.0 mL volumetric flask.

Step 3: Add about 500 mL of distilled water to the flask, and swirl to dissolve the $\text{NaOH}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) Will the chemical quantities in the procedure produce a solution that is $0.500\text{ M NaOH}(aq)$? Explain your reasoning.

(k) Before adding the $\text{NaOH}(s)$ to the volumetric flask, the student first rinsed the volumetric flask with distilled water, leaving some droplets of distilled water on the inside of the flask. As a result of this step, will the concentration of $\text{NaOH}(aq)$ be greater than, less than, or equal to 0.500 M ? Explain your answer.

(l) When making the solution above, the student notices that the temperature increases when dissolving

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in 1000. g each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

Solvent	X	Y
Specific Heat Capacity (J/(g · °C))	2.91	2.53

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{BPB}][\text{H}_3\text{O}^+]}{[\text{HBPB}^+]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

· 800 nm. Photons with a wavelength of 800 nm have a lower frequency than those with a wavelength of 590 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the lower frequency (shorter wavelength) photons have the lower energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

· Endothermic. The diagram shows that the absorbance, and therefore the concentration, of BPB is greater at high temperature. The increased concentration of products at elevated temperature is consistent with an endothermic reaction, according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· 0.30

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly describes the effect of the error.

· The estimated molarity of BPB would be too low. The absorbance maximum of BPB is at 590 nm, so setting the spectrophotometer to any other wavelength will result in a lower absorbance even if the concentration is unchanged. This lower absorbance would be misinterpreted as a lower concentration of BPB than the actual concentration.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- ☐ The response indicates a value of 240 for the half-life
- ☐ The response has units of “s” or “seconds”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OH^- decreases, the collisions between reactant particles occur less frequently per unit of time. The lower frequency of collisions leads to a decreased rate of product formation and correspondingly a longer half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The activation energy is the same. Activation energy does not change with concentration. (Only a catalyst or different reaction pathway could alter the activation energy.)

Part J

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid explanation based on the chemical quantities.

· Yes. The student has dissolved 20.0 g (0.500 mol) NaOH in 1000.0 mL (1.000 L) of solution, resulting in a concentration of 0.500 *M*.

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The concentration of the NaOH solution will be equal to 0.500 *M*. The droplets of distilled water simply mix with the rest of the distilled water added in step 3. As long as the volumetric flask is filled to the correct mark in step 4, the concentration of the solution will be 0.500 *M*.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



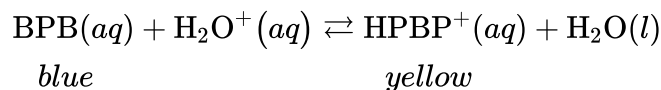
0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The temperature change would be less in solvent X than in solvent Y. Because X has the greater heat capacity, the same energy transferred to an equal mass of it will result in a smaller temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

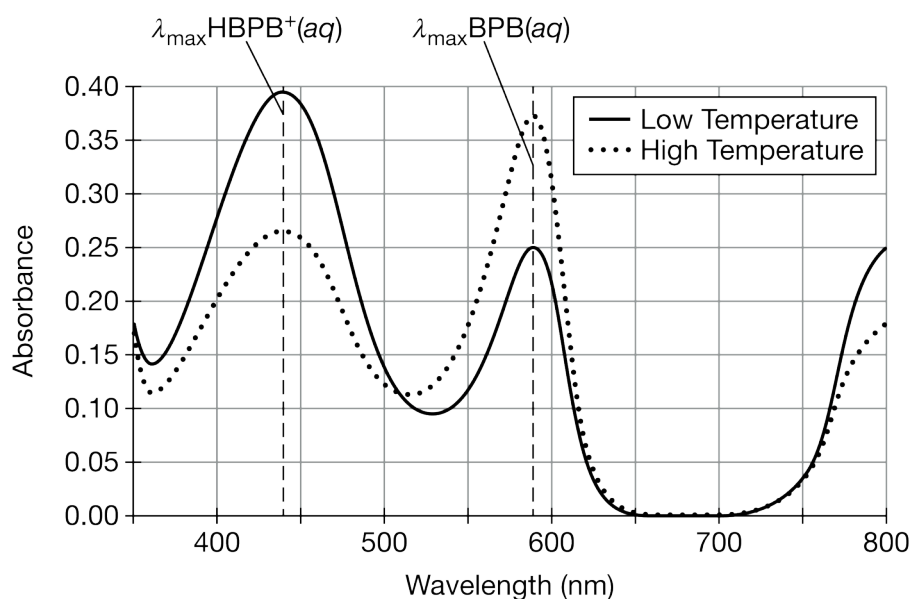
7. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Bromophenol blue, BPB, is a blue dye whose spectrum has a strong absorption peak at a wavelength of 590 nm. BPB reacts with hydronium ions according to the equation above, producing water and HPBP^+ , which is yellow and has a strong absorption peak at a wavelength of 440 nm.

- (a) Write the K_{eq} expression for the equilibrium reaction above.

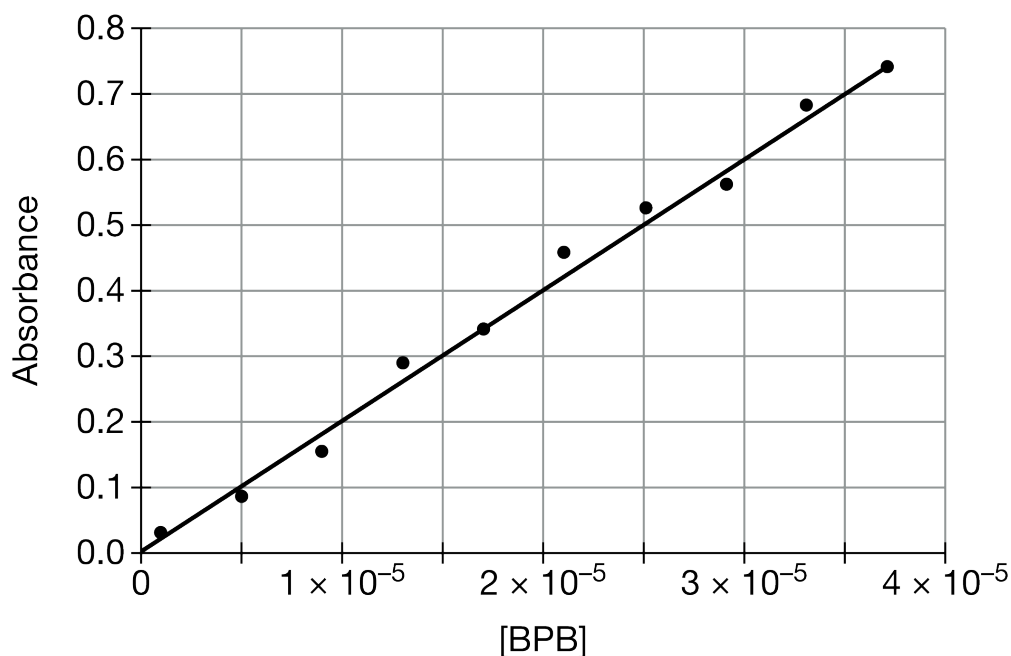
The absorption spectra of an equilibrium mixture of $\text{BPB}(aq)$ and $\text{HPBP}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 440 nm or the one at 590 nm, results from the absorption of photons with the greater energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 590 nm versus the concentration of BPB is shown in the graph below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line shown in the graph, what is the approximate absorbance of a solution in which $[\text{BPB}] = 3.5 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a BPB solution of unknown concentration in order to determine the $[\text{BPB}]$ in the solution using the calibration plot above. If a fingerprint remains on the cuvette and in the path of light in the spectrophotometer during the measurement, what would be the effect on the estimated molarity of BPB? Explain your reasoning.

Over time, BPB converts to a colorless molecule in the presence of OH^- ions. To study the rate of the reaction, the student recorded the absorbance of the BPB solution at 590 nm in a 0.50 M NaOH solution. The temperature and $[\text{OH}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (min)	Absorbance
0	0.60
4	0.48
8	0.38
12	0.30
16	0.24
20	0.19
24	0.15

(f) Determine the half-life of the reaction. Include units with your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to BPB.

A second trial is conducted at the same constant temperature, but with $[\text{OH}^-]$ changed to 2.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OH}^-]$ was 0.50 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OH}^-]$ is changed from 0.50 M to 2.00 M .

(i) Is the activation energy for the reaction when $[\text{OH}^-]$ is 2.00 M greater than, the same as, or less than the activation energy when $[\text{OH}^-]$ is 0.50 M ? Justify your answer.

In order to prepare the 2.00 M solution of $\text{NaOH}(aq)$, one student follows a procedure with the following steps.

Step 1: Measure out 16.0 g of $\text{NaOH}(s)$.

Step 2: Add the $\text{NaOH}(s)$ to a 100.0 mL volumetric flask that is half-filled with distilled water.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

(j) The chemical quantities used in the procedure do NOT produce $2.00\text{ M NaOH}(aq)$. Describe specifically how they should be changed in order to produce $2.00\text{ M NaOH}(aq)$.

(k) Another student is preparing a 2.00 M NaOH solution using the correct chemical quantities, but notices that some undissolved solid remains at the bottom of the volumetric flask. Is the concentration of $\text{NaOH}(aq)$ in the solution in the flask greater than, less than, or equal to 2.00 M ? Explain your answer.

(l) When preparing solutions of NaOH , the student notices that the temperature increases when dissolving $\text{NaOH}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{NaOH}(s)$ in $1000.\text{ g}$ each of two different solvents, X and water. Assuming the magnitude of the enthalpy change is the same regardless of solvent, would the temperature change in X be greater than, less than, or equal to the temperature change in the

Chapter 10 Part 2: Le Chatelier's Principle FRQ

water? Justify your answer. The specific heat capacities of water and solvent X are given in the table below.

Solvent	Water	X
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	4.18	2.53

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{HBPB}^+]}{[\text{BPB}][\text{H}_3\text{O}^+]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· 440 nm. Photons with a wavelength of 440 nm have a higher frequency than those with a wavelength of 590 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the higher frequency (shorter wavelength) photons have the greater energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· Exothermic. The diagram shows that the absorbance, and therefore the concentration, of BPB is greater at high temperature. The increased concentration of reactants at elevated temperature is consistent with an exothermic reaction,

Chapter 10 Part 2: Le Chatelier's Principle FRQ

according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· 0.70

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly describes the effect of the error.

· The estimated molarity of BPB would be too high. The fingerprint would scatter or absorb some of the incoming light, making the measured absorbance greater than it should be. Because absorbance and concentration are directly proportional, the greater absorbance would be misinterpreted as a concentration of BPB that is higher than the actual concentration.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

- ☐ The response indicates a value of 12 for the half-life
- ☐ The response has units of “min”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OH^- increases, the collisions between reactant particles occur more frequently per unit of time. The greater frequency of collisions leads to an increased rate of product formation and correspondingly a shorter half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The activation energy is the same. Activation energy does not change with concentration. (Only a catalyst or different reaction pathway could alter the activation energy.)

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides a valid correction to the chemical quantities.

- To make a 2.00 *M* solution, the student needs to dissolve 80.0 g of NaOH (2.00 mol) in 1.00 L of solution.

OR

- To make a 2.00 *M* solution, the student needs to dissolve 8.00 g of NaOH (0.200 mol) in 100.0 mL of solution.

OR

- This procedure makes a 4.00 *M* solution, so the student needs to dilute it by a factor of 2.

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

- Less than 2.00 *M*. There are fewer moles of NaOH dissolved in the solution than was intended.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

- The temperature change would be greater in solvent X than in water. Because X has a smaller heat capacity, the same energy transferred to an equal mass of it will result in a greater temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

8. The compound butane, C_4H_{10} , occurs in two isomeric forms, *n*-butane and isobutane (2-methyl propane). Both compounds exist as gases at 25°C and 1.0 atm.

- Draw the structural formula of each of the isomers (include all atoms). Clearly label each structure.
- On the basis of molecular structure, identify the isomer that has the higher boiling point. Justify your answer.

The two isomers exist in equilibrium as represented by the equation below.



Suppose that a 0.010 mol sample of pure *n*-butane is placed in an evacuated 1.0 L rigid container at 25°C .

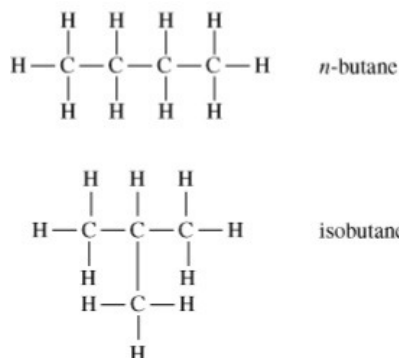
- Write the expression for the equilibrium constant, K_c , for the reaction.
- Calculate the initial pressure in the container when the *n*-butane is first introduced (before the reaction starts).
- The *n*-butane reacts until equilibrium has been established at 25°C .
 - Calculate the total pressure in the container at equilibrium. Justify your answer.
 - Calculate the molar concentration of each species at equilibrium.
 - If the volume of the system is reduced to half of its original volume, what will be the new concentration of *n*-butane after equilibrium has been reestablished at 25°C ? Justify your answer.

Suppose that in another experiment a 0.010 mol sample of pure isobutane is placed in an evacuated 1.0 L rigid container and allowed to come to equilibrium at 25°C .

- Calculate the molar concentration of each species after equilibrium has been established.

Part A

2 points are earned for two correct structures with correct labels



Chapter 10 Part 2: Le Chatelier's Principle FRQ

(Note: 1 point can be earned for either two correct structures that are mislabeled or one correct structure with or without correct label.)

OR

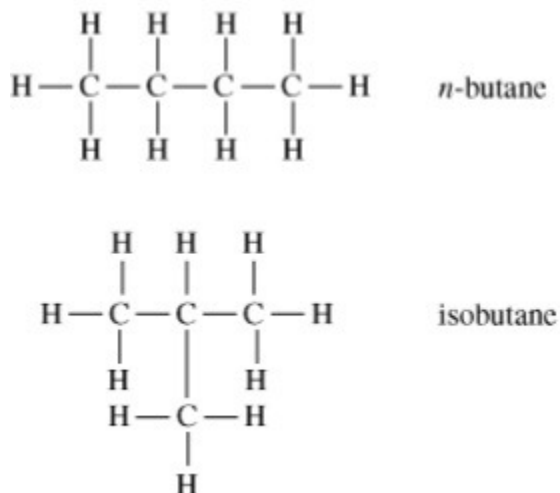
1 point can be earned for two skeletal structures (hydrogen atoms not shown) with proper labels.



0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for two correct structures with correct labels



(Note: 1 point can be earned for either two correct structures that are mislabeled or one correct structure with or without correct label.)

OR

1 point can be earned for two skeletal structures (hydrogen atoms not shown) with proper labels.

Part B

1 point is earned for the correct choice of isomer with justification.

The isomer *n*-butane has the higher boiling point. London (dispersion) forces are greater among molecules of *n*-butane than they are among molecules of isobutane because molecules of *n*-butane, with its linear structure, can approach one another more closely and can form a greater number of induced temporary dipoles than molecules of isobutane, with its more compact structure, can form.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct choice of isomer with justification.

The isomer *n*-butane has the higher boiling point. London (dispersion) forces are greater among molecules of *n*-butane than they are among molecules of isobutane because molecules of *n*-butane, with its linear structure, can approach one another more closely and can form a greater number of induced temporary dipoles than molecules of isobutane, with its more compact structure, can form.

Part C

1 point is earned for the correct equation.

$$K_c = \frac{[\text{isobutane}]}{[\text{n-butane}]}$$

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct equation.

$$K_c = \frac{[\text{isobutane}]}{[\text{n-butane}]}$$

Part D

1 point is earned for the correct substitution and numerical answer.

$$P = \frac{nRT}{V} = \frac{(0.010 \text{ mol})(0.0821 \frac{\text{L} \times \text{atm}}{\text{mol} \times \text{K}})(298 \text{ K})}{1.0 \text{ L}}$$

$$= 0.24 \text{ atm}$$

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct substitution and numerical answer.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$P = \frac{nRT}{V} = \frac{(0.010 \text{ mol})(0.0821 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}})(298 \text{ K})}{1.0 \text{ L}}$$

$$= 0.24 \text{ atm}$$

Part E

- 1 point is earned for the correct answer with justification.

The total pressure in the container remains the same, 0.24 atm. As the reaction proceeds, the number of molecules in the container remains constant; one molecule of isobutane is produced for each molecule of *n*-butane consumed.

- 1 point is earned for the correct setup.

$$K_c = \frac{[\text{isobutane}]}{[\text{n-butane}]} = \frac{x}{(0.010-x)} = 2.5$$

1 point is earned for both correct numerical answers.

$$x = 2.5 (0.010 - x) = 0.025 - 2.5x$$

$$3.5x = 0.025 \Rightarrow x = 0.0071 \text{ M isobutane}$$

$$(0.010 \text{ M} - 0.0071 \text{ M}) = 0.003 \text{ M } n\text{-butane}$$

- 1 point is earned for the correct answer with justification.

Halving the volume of the container at equilibrium doubles the pressure of both isobutane and *n*-butane, which has no effect on the equilibrium because the stoichiometry of the reaction is one mole of product produced for each mole of reactant consumed. Since the number of moles of each isomer is unchanged but the volume is reduced by half, concentrations of both isomers are doubled and the concentration of *n*-butane will be $2 \times 0.003 \text{ M} = 0.006 \text{ M}$.



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

- 1 point is earned for the correct answer with justification.

The total pressure in the container remains the same, 0.24 atm. As the reaction proceeds, the number of molecules in the container remains constant; one molecule of isobutane is produced for each molecule of *n*-butane consumed.

- 1 point is earned for the correct setup.

$$K_c = \frac{[\text{isobutane}]}{[\text{n-butane}]} = \frac{x}{(0.010-x)} = 2.5$$

1 point is earned for both correct numerical answers.

$$x = 2.5 (0.010 - x) = 0.025 - 2.5x$$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$3.5x = 0.025 \Rightarrow x = 0.0071 \text{ M isobutane}$$

$$(0.010 \text{ M} - 0.0071 \text{ M}) = 0.003 \text{ M } n\text{-butane}$$

- 1 point is earned for the correct answer with justification.

Halving the volume of the container at equilibrium doubles the pressure of both isobutane and *n*-butane, which has no effect on the equilibrium because the stoichiometry of the reaction is one mole of product produced for each mole of reactant consumed. Since the number of moles of each isomer is unchanged but the volume is reduced by half, concentrations of both isomers are doubled and the concentration of *n*-butane will be $2 \times 0.003 \text{ M} = 0.006 \text{ M}$.

Part F

1 point is earned for correct numerical answers or a correct statement regarding their equivalence to values obtained in part (e)(ii).

The concentrations of isobutane and *n*-butane would be the same as they were calculated in part (e)(ii), 0.0071 *M* and 0.003 *M*, respectively.



0	1
---	---

The student response earns one of the following points:

1 point is earned for correct numerical answers or a correct statement regarding their equivalence to values obtained in part (e)(ii).

The concentrations of isobutane and *n*-butane would be the same as they were calculated in part (e)(ii), 0.0071 *M* and 0.003 *M*, respectively.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

9. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about $\text{C}_2\text{H}_4\text{O}$.

- (a) $\text{C}_2\text{H}_4\text{O}$ (molar mass 44.06g/mol) is a gas at room temperature and can be harmful at concentrations above $8.17 \times 10^{-6} M$. What is the maximum mass of this compound that can safely be present in a room with a volume of $3.00 \times 10^5 \text{L}$?

Two structures, X and Z, with the molecular formula $\text{C}_2\text{H}_4\text{O}$ are shown in the table.

Structure	Lewis Dot Diagram
X	
Z	

- (b) Based on VSEPR theory, predict the $\text{H} - \text{C} - \text{H}$ bond angle in structure X.
- (c) A student incorrectly claims that X and Z are resonance structures of each other. Explain why the student's claim is incorrect.
- X and Z are both gases at room temperature and 1atm , and they react according to the following equation.
- $$\text{X} \rightleftharpoons \text{Z}$$
- (d) Write the expression for the equilibrium constant, K_p , in terms of the partial pressures of X and Z.
- (e) For the reaction represented above, the value of the equilibrium constant, K_p , is 3.3×10^{-8} at 25°C . Which would

Chapter 10 Part 2: Le Chatelier's Principle FRQ

have a greater partial pressure at equilibrium: the reactant, **X**, or the product, **Z**? Circle one of the options and justify your choice.

Option 1 :

The partial pressure of reactant **X** is greater.

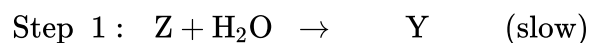
Option 2 :

The partial pressure of product **Z** is greater.

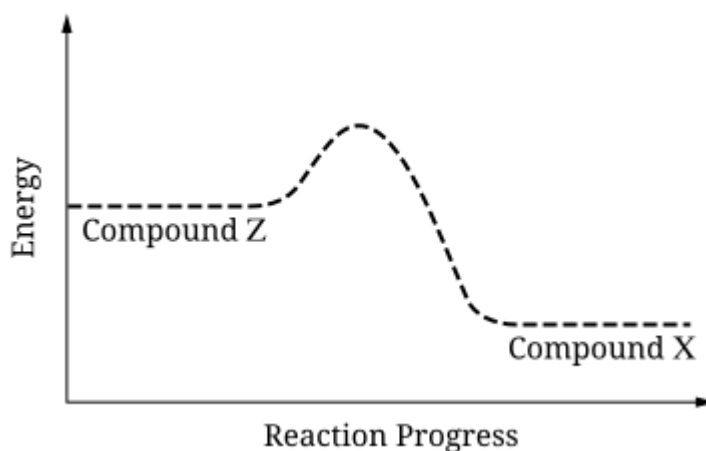
(f) An empty vessel is filled with compound **X** and sealed shut. The initial pressure of compound **X** is 0.55 atm. Calculate the partial pressure of compound **Z** in the vessel after equilibrium is established at constant temperature.

(g) The sample is compressed to half of the original volume, without any change in temperature. Does the value of K_p increase, decrease, or remain the same? Justify your choice.

(h) In the reverse reaction, the conversion of compound **Z** to compound **X** is catalyzed in the presence of trace amounts of water. The catalyzed reaction is proposed to occur via the following two-step mechanism.



(i) Identify the intermediate in the mechanism, if any.



(ii) A reaction profile of the uncatalyzed reaction is shown above. On the same axes, sketch a reaction profile consistent with the proposed mechanism for the catalyzed reaction.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides the correct calculated value.

- $(8.17 \times 10^{-6} M)(3.00 \times 10^5 L) = 2.451 \text{ mol}$
 $2.451 \text{ mol} \times \frac{44.06 \text{ g}}{1 \text{ mol}} = 108 \text{ g}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer.

- 109.5°

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation.

- In resonance structures, the connectivity of atoms must remain the same; only electron arrangements change. In the case of X and Z, a hydrogen atom in molecule X has been moved to a different position in molecule Z, so X and Z cannot be resonance structures of each other.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct expression.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$\bullet K_p = \frac{P_Z}{P_X}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

- Option 1 is circled.

The value of K_p is less than 1. At equilibrium, the partial pressure of the reactant is greater than the partial pressure of the product.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$\bullet K_p = \frac{P_Z}{P_X}$$

$$3.3 \times 10^{-8} = \frac{P_Z}{(0.55 \text{ atm} - P_Z)} \approx \frac{P_Z}{(0.55 \text{ atm})}$$

$$P_Z = (3.3 \times 10^{-8})(0.55 \text{ atm}) = 1.8 \times 10^{-8} \text{ atm}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct answer and a valid justification.

- K_p will remain the same. Only a temperature change can alter the value of K .

OR

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- K_p will remain the same. Changing the volume of the container does not change the K_p value.

Part H(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer.

- Y is an intermediate.

Part H(ii)

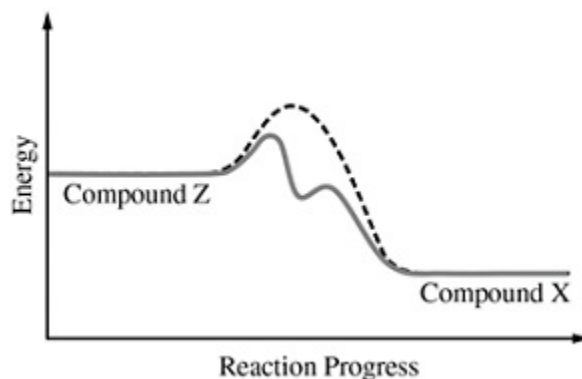
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

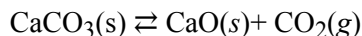
The response does not meet **any** of the following criteria.

- ☐ The response shows the correct starting and ending points.
 - The curve begins and ends at the same energies as the original curve, with an initial activation energy depicted.
- ☐ The response shows a curve with the correct shape.
 - The curve should have two local maxima, with the first step having a larger E_a than the second step (and smaller than the E_a of the uncatalyzed reaction).

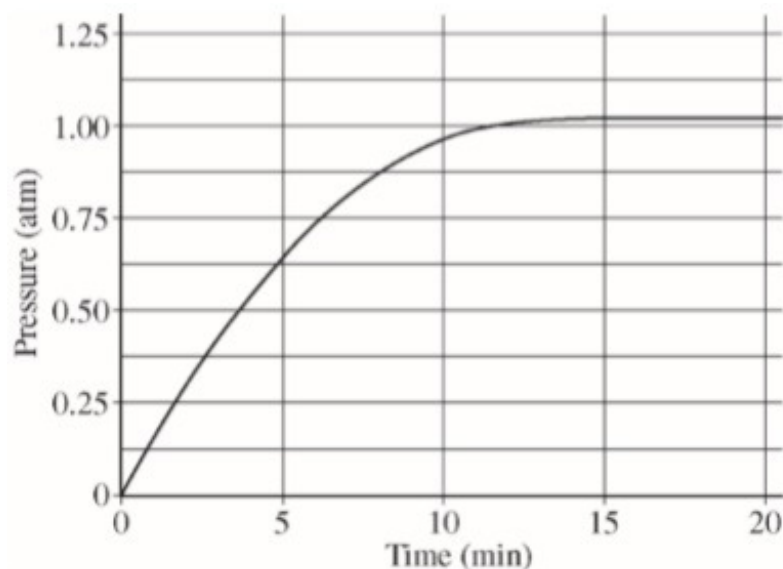


Chapter 10 Part 2: Le Chatelier's Principle FRQ

10.



When heated, calcium carbonate decomposes according to the equation above. In a study of the decomposition of calcium carbonate, a student added a 50.0 g sample of powdered $\text{CaCO}_3(s)$ to a 1.00 L rigid container. The student sealed the container, pumped out all the gases, then heated the container in an oven at 1100 K. As the container was heated, the total pressure of the $\text{CO}_2(g)$ in the container was measured over time. The data are plotted in the graph below.



The student repeated the experiment, but this time the student used a 100.0 g sample of powdered $\text{CaCO}_3(s)$. In this experiment, the final pressure in the container was 1.04 atm, which was the same final pressure as in the first experiment.

- Calculate the number of moles of $\text{CO}_2(g)$ present in the container after 20 minutes of heating.
- The student claimed that the final pressure in the container in each experiment became constant because all of the $\text{CaCO}_3(s)$ had decomposed. Based on the data in the experiments, do you agree with this claim? Explain.
- After 20 minutes some $\text{CO}_2(g)$ was injected into the container, initially raising the pressure to 1.5 atm. Would the final pressure inside the container be less than, greater than, or equal to 1.04 atm? Explain your reasoning.
- Are there sufficient data obtained in the experiments to determine the value of the equilibrium constant, K_p , for the decomposition of $\text{CaCO}_3(s)$ at 1100 K? Justify your answer.

Part A

1 point is earned for the proper setup using the ideal gas law and an answer that is consistent with the setup.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$PV = nRT$$

$$\frac{PV}{RT} = n = \frac{(1.04 \text{ atm})(1.00 \text{ L})}{(0.0821 \frac{\text{L atm}}{\text{mol K}})(1100 \text{ K})} = 0.0115 \text{ mol CO}_2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the proper setup using the ideal gas law and an answer that is consistent with the setup.

$$PV = nRT$$

$$\frac{PV}{RT} = n = \frac{(1.04 \text{ atm})(1.00 \text{ L})}{(0.0821 \frac{\text{L atm}}{\text{mol K}})(1100 \text{ K})} = 0.0115 \text{ mol CO}_2$$

Part B

1 point is earned for disagreement with the claim and for a correct justification using stoichiometry or a discussion of the creation of an equilibrium condition.

Do not agree with claim

Explanation I: In experiment 1, the moles of $\text{CaCO}_3 = 50.0 \text{ g}/100.09 \text{ g/mol} = 0.500 \text{ mol CaCO}_3$. If the reaction had gone to completion, 0.500 mol of CO_2 would have been produced. From part (a) only 0.0115 mol was produced. Hence, the student's claim was false.

Explanation II: The two different experiments (one with 50.0 g of CaCO_3 and one with 100.0 g of CaCO_3) reached the same constant, final pressure of 1.04 atm. Since increasing the amount of reactant did not produce more product, there is no way that all of the CaCO_3 reacted. Instead, an equilibrium condition has been achieved and there must be some solid CaCO_3 in the container.



0	1
---	---

The student response earns one of the following points:

1 point is earned for disagreement with the claim and for a correct justification using stoichiometry or a discussion of the creation of an equilibrium condition.

Do not agree with claim

Explanation I: In experiment 1, the moles of $\text{CaCO}_3 = 50.0 \text{ g}/100.09 \text{ g/mol} = 0.500 \text{ mol CaCO}_3$. If the reaction had gone to completion, 0.500 mol of CO_2 would have been produced. From part (a) only 0.0115 mol was produced. Hence, the student's claim was false.

Explanation II: The two different experiments (one with 50.0 g of CaCO_3 and one with 100.0 g of CaCO_3) reached the

Chapter 10 Part 2: Le Chatelier's Principle FRQ

same constant, final pressure of 1.04 atm. Since increasing the amount of reactant did not produce more product, there is no way that all of the CaCO_3 reacted. Instead, an equilibrium condition has been achieved and there must be some solid CaCO_3 in the container.

Part C

1 point is earned for the correct answer with justification.

The final pressure would be equal to 1.04 atm. Equilibrium was reached in both experiments; the equilibrium pressure at this temperature is 1.04 atm. As the reaction shifts toward the reactant, the amount of $\text{CO}_2(\text{g})$ in the container will decrease until the pressure returns to 1.04 atm.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

The final pressure would be equal to 1.04 atm. Equilibrium was reached in both experiments; the equilibrium pressure at this temperature is 1.04 atm. As the reaction shifts toward the reactant, the amount of $\text{CO}_2(\text{g})$ in the container will decrease until the pressure returns to 1.04 atm.

Part D

1 point is earned for correct explanation that is consistent with the student's answer to part (b).

Yes. For the equilibrium reaction represented by the chemical equation in this problem, at a given temperature the equilibrium pressure of CO_2 determines the equilibrium constant. Since the measured pressure of CO_2 is also the equilibrium pressure of CO_2 , $K_p = P_{\text{CO}_2} = 1.04$.

Note: If the response in part (b) indicates "yes", that all of the $\text{CaCO}_3(\text{s})$ had decomposed, then the point can be earned by stating that the system did not reach equilibrium in either experiment and hence the value of K_p cannot be calculated from the data.



0	1
---	---

The student response earns one of the following points:

1 point is earned for correct explanation that is consistent with the student's answer to part (b).

Yes. For the equilibrium reaction represented by the chemical equation in this problem, at a given temperature the equilibrium pressure of CO_2 determines the equilibrium constant. Since the measured pressure of CO_2 is also the equilibrium pressure of CO_2 , $K_p = P_{\text{CO}_2} = 1.04$.

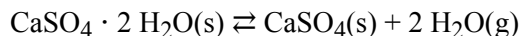
Note: If the response in part (b) indicates "yes", that all of the $\text{CaCO}_3(\text{s})$ had decomposed, then the point can be earned by stating that the system did not reach equilibrium in either experiment and hence the value of K_p cannot be calculated from

Chapter 10 Part 2: Le Chatelier's Principle FRQ

the data.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

11.



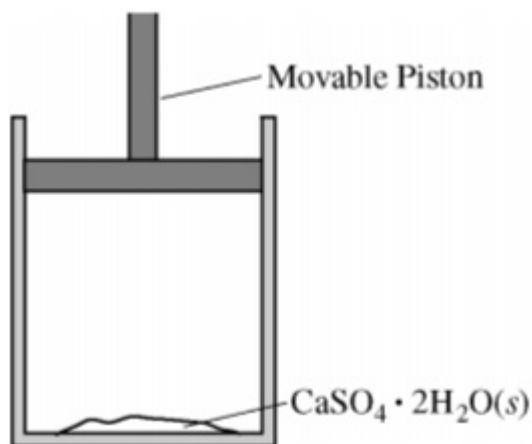
The hydrate $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ can be heated to form the anhydrous salt, $\text{CaSO}_4(s)$, as shown by the reaction represented above.

- a. Using the data in the table below, calculate the value of ΔG° , in kJ/mol_{rxn} , for the reaction at 298 K.

Substance	ΔG_f° at 298 K (kJ/mol)
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$	-1795.70
$\text{CaSO}_4(s)$	-1320.30
$\text{H}_2\text{O}(g)$	-228.59

- b. Given that the value of ΔH° for the reaction at 298 K is $+105 \text{ kJ/mol}_{rxn}$, calculate the value of ΔS° for the reaction at 298 K. Include units with your answer.

A sample of $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ is placed in a cylinder with a movable piston as shown in the diagram below. The air above the solid is at 1.00 atm and is initially dry (partial pressure of $\text{H}_2\text{O}(g) = 0 \text{ atm}$).



- c. Write the expression for the equilibrium constant, K_p , for the reaction.
- d. Given that the equilibrium constant, K_p , is 6.4×10^{-4} at 298 K, determine the partial pressure, in atm, of water vapor in the cylinder at equilibrium at 298 K.
- e. If the volume of the system is reduced to one-half of its original volume and the system is allowed to reestablish equilibrium at 298 K, what will be the pressure, in atm, of the water vapor at the new volume? Justify your answer.

In the laboratory, the hydrate $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ can be heated in a crucible to completely drive off the water of hydration to form the anhydrous salt, $\text{CaSO}_4(s)$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- f. A 2.49 g sample of pure $\text{CaSO}_4 \cdot 2 \text{H}_2\text{O}(s)$ is heated several times until the mass is constant. Calculate the mass, in grams, of the solid that remains after the dehydration reaction is complete.

Part A

1 point is earned for the mole factor for water.

$$\Delta G^\circ = \Sigma \Delta G^\circ_f \text{ products} - \Sigma \Delta G^\circ_f \text{ reactants}$$

$$= -1320.30 + (2 \times -228.59) - (-1795.70 \text{ kJ mol}^{-1})$$

1 point is earned for answer (units are optional)

$$= 18.22 \text{ kJ mol}^{-1}$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the mole factor for water.

$$\Delta G^\circ = \Sigma \Delta G^\circ_f \text{ products} - \Sigma \Delta G^\circ_f \text{ reactants}$$

$$= -1320.30 + (2 \times -228.59) - (-1795.70 \text{ kJ mol}^{-1})$$

1 point is earned for answer (units are optional)

$$= 18.22 \text{ kJ mol}^{-1}$$

Part B

1 point is earned for the correct substitution of ΔG° , ΔH° , and T.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

1 point is earned for the correct answer *with correct units*.

$$\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{(105 - 18.22) \text{ kJ mol}^{-1}}{298 \text{ K}}$$

$$= 0.29 \text{ kJ K}^{-1} \text{ mol}^{-1} \text{ OR } 290 \text{ J K}^{-1} \text{ mol}^{-1}$$



0	1	2
---	---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The student response earns two of the following points:

1 point is earned for the correct substitution of ΔG° , ΔH° , and T.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

1 point is earned for the correct answer *with correct units*.

$$\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{(105 - 18.22) \text{ kJ mol}^{-1}}{298 \text{ K}}$$

$$= 0.29 \text{ kJ K}^{-1} \text{ mol}^{-1} \text{ OR } 290 \text{ J K}^{-1} \text{ mol}^{-1}$$

Part C

1 point is earned for the correct expression (use of partial pressure only).

$$K_p = (p_{\text{H}_2\text{O}})^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct expression (use of partial pressure only).

$$K_p = (p_{\text{H}_2\text{O}})^2$$

Part D

1 point is earned for the correct partial pressure (units are not required).

$$K_p = (p_{\text{H}_2\text{O}})^2 \Rightarrow 6.4 \times 10^{-4}$$

$$p_{\text{H}_2\text{O}} = \sqrt{6.4 \times 10^{-4}} = 0.025 \text{ atm}$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct partial pressure (units are not required).

$$K_p = (p_{\text{H}_2\text{O}})^2 \Rightarrow 6.4 \times 10^{-4}$$

$$p_{\text{H}_2\text{O}} = \sqrt{6.4 \times 10^{-4}} = 0.025 \text{ atm}$$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Part E

1 point is earned for the correct answer with justification.

The $p_{\text{H}_2\text{O}}$ at equilibrium at the new volume will be 0.025 atm. Equilibrium vapor pressure is dependent on K_p , which in turn is a function of temperature, not volume. Because the temperature is still 298 K, the vapor pressure of H_2O remains 0.025 atm in the new volume.



0	1
---	---

The student response earns one of the following points:

1 point is earned for the correct answer with justification.

The $p_{\text{H}_2\text{O}}$ at equilibrium at the new volume will be 0.025 atm. Equilibrium vapor pressure is dependent on K_p , which in turn is a function of temperature, not volume. Because the temperature is still 298 K, the vapor pressure of H_2O remains 0.025 atm in the new volume.

Part F

1 point is earned for the (rounded) correct molar masses.

molar mass of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s) = 172.172 \text{ g mol}^{-1}$

molar mass of $\text{CaSO}_4(s) = 136.14 \text{ g mol}^{-1}$

1 point is earned for an answer consistent with the molar masses.

$$2.49 \text{ g CaSO}_4 \cdot 2\text{H}_2\text{O} \times \frac{136.14 \text{ g CaSO}_4}{172.172 \text{ g CaSO}_4 \cdot 2\text{H}_2\text{O}} = 1.97 \text{ g CaSO}_4$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the (rounded) correct molar masses.

molar mass of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s) = 172.172 \text{ g mol}^{-1}$

molar mass of $\text{CaSO}_4(s) = 136.14 \text{ g mol}^{-1}$

1 point is earned for an answer consistent with the molar masses.

$$2.49 \text{ g CaSO}_4 \cdot 2\text{H}_2\text{O} \times \frac{136.14 \text{ g CaSO}_4}{172.172 \text{ g CaSO}_4 \cdot 2\text{H}_2\text{O}} = 1.97 \text{ g CaSO}_4$$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

12. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{Cl}_2(g)$ and $\text{CO}(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



- (a) If 7.0 g of $\text{CO}(g)$ is consumed in the reaction with excess $\text{Cl}_2(g)$, how many moles of $\text{COCl}_2(g)$ are produced?
- (b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

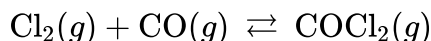
Time	P_{Cl_2} (atm)	P_{CO} (atm)	P_{COCl_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.22
t_2	0.62	0.62	0.38
t_3	0.51	0.51	0.49
t_4	0.45	0.45	0.55
t_5	0.42	0.42	0.58
t_6	0.41	0.41	0.59
t_7	0.40	0.40	0.60
t_8	0.40	0.40	0.60
t_9	0.40	0.40	0.60

- (c) At time t_8 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds the most molecules of which gas, $\text{Cl}_2(g)$ or $\text{COCl}_2(g)$? Explain your answer.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of Cl_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Cl}_2(g)$ or $\text{COCl}_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of the number of moles of COCl_2 .

$$7.0 \text{ g CO} \times \frac{1 \text{ mol CO}}{28 \text{ g CO}} = 0.25 \text{ mol CO}$$

$$0.25 \text{ mol CO} \times \frac{1 \text{ mol COCl}_2}{1 \text{ mol CO}} = 0.25 \text{ mol COCl}_2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the element that is oxidized and a correct justification equivalent to the following.

· Carbon. Its oxidation number changes from +2 to +4.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides the correct answer and a valid justification equivalent to the following.

· Equal to. The partial pressure of the product (or reactants) remains the same from time t_8 to t_9 , indicating that the reaction has reached equilibrium.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· COCl_2 . At equilibrium (t_7 , t_8 , t_9) the partial pressure of COCl_2 is greatest and thus the number of molecules of COCl_2 is greatest.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a valid justification equivalent to one of the following.

· Disagree. According to Le Châtelier's principle, the distribution of species will shift towards products when the temperature is decreased in an exothermic reaction. Therefore, the mole fraction of Cl_2 will decrease, not increase.

OR

· Agree. Decreasing the temperature of the constant-volume system would decrease the pressure in the container. According to Le Châtelier's principle, the distribution of species will shift towards the side with the greater number of moles. Therefore, the mole fraction of Cl_2 will increase.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response shows the correct calculation of K_p .

$$K_p = \frac{P_{\text{COCl}_2}}{(P_{\text{Cl}_2})(P_{\text{CO}})} = \frac{0.60}{(0.40)(0.40)} = 3.75 \approx 3.8$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows **one** of the following.

- ☐ $K_p' = \frac{(P_{\text{COCl}_2})^3}{(P_{\text{Cl}_2})^3(P_{\text{CO}})^3} = \frac{(0.60)^3}{(0.40)^3(0.40)^3} = 53$
- ☐ $K_p' = (K_p)^3 = (3.8)^3 = 55$ (53 if not rounded after part (f))

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

$\text{COCl}_2(g)$, is more likely to deviate from the ideal gas law. $\text{Cl}_2(g)$ has dispersion forces only, while $\text{COCl}_2(g)$ has dipole-dipole attractions in addition to stronger dispersion forces than $\text{Cl}_2(g)$ due to a larger and more polarizable electron cloud in $\text{COCl}_2(g)$. Therefore, $\text{COCl}_2(g)$ has stronger overall intermolecular forces.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

13. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{Cl}_2(g)$ and $\text{O}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



- (a) If 6.4 g of $\text{O}_2(g)$ is consumed in the reaction with excess $\text{Cl}_2(g)$, how many moles of $\text{Cl}_2\text{O}(g)$ are produced?
- (b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

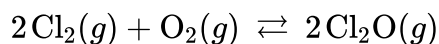
The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{Cl_2} (atm)	P_{O_2} (atm)	$P_{\text{Cl}_2\text{O}}$ (atm)
t_0	0.50	0.50	0.0
t_1	0.42	0.46	0.08
t_2	0.36	0.43	0.14
t_3	0.32	0.41	0.19
t_4	0.29	0.39	0.22
t_5	0.27	0.38	0.23
t_6	0.26	0.37	0.24
t_7	0.25	0.38	0.25
t_8	0.25	0.38	0.25
t_9	0.25	0.38	0.25

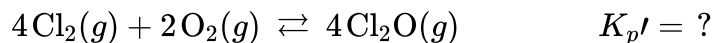
- (c) At time t_4 , is the rate of the reverse reaction greater than, less than, or equal to the rate of the forward reaction? Justify your choice.
- (d) At equilibrium, the container holds fewer molecules of which gas, $\text{Cl}_2(g)$ or $\text{O}_2(g)$? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of $\text{Cl}_2\text{O}(g)$ in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by

Chapter 10 Part 2: Le Chatelier's Principle FRQ

the equation below.



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Cl}_2(g)$ or $\text{O}_2(g)$, will deviate most from the ideal gas law at low temperature? Justify your choice.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of the number of moles of Cl_2O .

$$6.4 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32 \text{ g O}_2} = 0.20 \text{ mol O}_2$$

$$0.20 \text{ mol O}_2 \times \frac{2 \text{ mol Cl}_2\text{O}}{1 \text{ mol O}_2} = 0.40 \text{ mol Cl}_2\text{O}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the element that is oxidized and a valid justification equivalent to the following.

· Chlorine. Its oxidation number changes from 0 to +1.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Less than. The partial pressures of the reactants decrease from time t_4 to time t_5 , indicating that the reverse reaction is occurring at the lesser rate.

OR

· Less than. The partial pressure of Cl_2O continues to increase after time t_4 , indicating that the forward reaction is proceeding at a greater rate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and an explanation equivalent to the following.

· Cl_2 . At equilibrium (t_7 , t_8 , t_9) the partial pressure of Cl_2 is lower than that of O_2 , indicating that there are fewer molecules of Cl_2 .

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. According to Le Châtelier's principle, the distribution of species will shift towards reactants when the temperature is decreased in an endothermic reaction. Therefore, the mole fraction of Cl_2O will decrease, not increase.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of K_p .

$$K_p = \frac{(P_{\text{Cl}_2\text{O}})^2}{(P_{\text{Cl}_2})^2(P_{\text{O}_2})} = \frac{(0.25)^2}{(0.25)^2(0.38)} = 2.6$$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows **one** of the following.

- ☐ $K_p' = \frac{(P_{\text{Cl}_2\text{O}})^4}{(P_{\text{Cl}_2})^4(P_{\text{O}_2})^2} = \frac{(0.25)^4}{(0.25)^4(0.38)^2} = 6.9$
- ☐ $K_p' = (K_p)^2 = (2.6)^2 = 6.8$ (6.9 if not rounded after part (f))

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



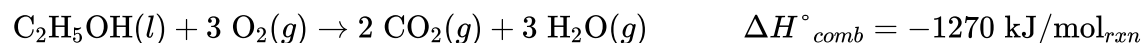
0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· $\text{Cl}_2(g)$ is more likely to deviate from the ideal gas law. $\text{Cl}_2(g)$ molecules have larger and more polarizable electron clouds than $\text{O}_2(g)$. Therefore, $\text{Cl}_2(g)$ has stronger overall intermolecular forces.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

14. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Ethanol, $\text{C}_2\text{H}_5\text{OH}$, will combust in air according to the equation above.

- (a) Is $\text{O}_2(g)$ oxidized in the reaction, or is it reduced? Justify your answer in terms of oxidation numbers.
- (b) When a sample of $\text{C}_2\text{H}_5\text{OH}$ was combusted, the volume of $\text{CO}_2(g)$ produced was 18.0 L when measured at 21.7°C and 1.03 atm. Determine the number of moles of $\text{CO}_2(g)$ that was produced.
- (c) Determine the volume of $\text{C}_2\text{H}_5\text{OH}(l)$, in mL, that was combusted to produce the volume of $\text{CO}_2(g)$ collected in part (b). (The density of $\text{C}_2\text{H}_5\text{OH}(l)$ is 0.79 g/mL.)
- (d) Determine the amount of heat, in kJ, that is released by the combustion reaction.

The combustion reaction occurred in a closed room containing 5.56×10^4 g of air originally at 21.7°C . Assume that all of the heat produced by the reaction was absorbed by the air (specific heat = $1.005 \text{ J/(g}^\circ\text{C)}$) in the room.

- (e) Determine the final temperature of the air in the room after the combustion.
- (f) The boiling points of ethanol and dimethyl ether are provided in the following table. Identify the intermolecular force that is most responsible for the difference in boiling point. In terms of this force, explain why the boiling point of ethanol is higher than that of dimethyl ether.

Compound	Structural Formula	Boiling Point ($^\circ\text{C}$)
Ethanol	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	78
Dimethyl ether	$\begin{array}{c} \text{H} \quad \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$	-24

One method of producing ethanol is by the reaction of ethene and water, as represented below.



Chapter 10 Part 2: Le Chatelier's Principle FRQ

(g) A chemist wants to run the reaction and maximize the amount of $\text{C}_2\text{H}_5\text{OH}(g)$ produced. Identify two ways the chemist could change the reaction conditions (other than adding or removing any chemical species) to favor the formation of more product. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets the following criteria.

- ☐ The response indicates that O_2 is reduced.
- ☐ The response indicates that the oxidation number of oxygen changes from zero in O_2 to -2 in CO_2 and H_2O .

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives a calculation equivalent to the following.

$$n = \frac{PV}{RT} = \frac{(1.03 \text{ atm})(18.0 \text{ L})}{(0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1})(273.15 + 21.7 \text{ K})} = 0.766 \text{ mol CO}_2$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If an incorrect answer to part (b) is used correctly in part (c), full credit can be earned in part (c).



0	1	2
---	---	---

The response meets both of the following criteria.

- ☐ The response indicates that 1 mol of $\text{C}_2\text{H}_5\text{OH}$ produces 2 mol of CO_2 .
- ☐ The response gives a calculation equivalent to the following.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$0.766 \text{ mol CO}_2 \times \frac{1 \text{ mol C}_2\text{H}_5\text{OH}}{2 \text{ mol CO}_2} \times \frac{46.07 \text{ g C}_2\text{H}_5\text{OH}}{1 \text{ mol C}_2\text{H}_5\text{OH}} \times \frac{1 \text{ mL}}{0.79 \text{ g}} = 22 \text{ mL}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If an incorrect answer to part (b) is used correctly in part (d), full credit can be earned in part (d).



0	1
---	---

The response gives a calculation equivalent to the following.

$$q = n \Delta H_{comb}^\circ$$

$$q = 0.766 \text{ mol CO}_2 \times \frac{1 \text{ mol}_{rxn}}{2 \text{ mol CO}_2} \times \frac{-1270 \text{ kJ}}{1 \text{ mol}_{rxn}} = -486 \text{ kJ}$$

The amount of heat released is 486 kJ.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If an incorrect answer to part (d) is used correctly in part (e), full credit can be earned in part (e).



0	1
---	---

The response gives a calculation equivalent to the following.

$$q_{air} = -q_{rxn} = +486 \text{ kJ} = +486,000 \text{ J}$$

$$q = mc\Delta T \Rightarrow \Delta T = \frac{q}{mc} = \frac{486,000 \text{ J}}{(5.56 \times 10^4 \text{ g})(1.005 \text{ J/(g } ^\circ\text{C)})} = 8.70^\circ\text{C}$$

$$T_{final} = 21.7^\circ\text{C} + 8.70^\circ\text{C} = 30.4^\circ\text{C}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets both of the following criteria.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- ☐ The response indicates that hydrogen bonding is the intermolecular force most responsible for the difference in boiling point.
- ☐ The response links the greater strength of the intermolecular forces to a higher boiling point (e.g., by stating that ethanol can form hydrogen bonds between its molecules, whereas dimethyl ether cannot, and that the attractions between molecules are stronger in ethanol than in dimethyl ether and therefore the boiling point of ethanol is higher).

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets both of the following criteria.

- ☐ The response indicates that the chemist could decrease the volume of the reaction container, and that this would work because there are fewer moles of gaseous product than moles of gaseous reactants; therefore, an increase in pressure due to a decrease in volume would favor the formation of product.
- ☐ The response indicates that the chemist could lower the temperature, and that this would work because a lower temperature favors an exothermic reaction, leading to the conversion of reactants into product.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

15. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{F}_2(g)$ and $\text{O}_2(g)$ are placed in a previously evacuated rigid container, where they react according to the equation below.



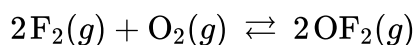
- (a) If 3.8 g of $\text{F}_2(g)$ is consumed in the reaction, how many moles of $\text{O}_2(g)$ will be needed?
- (b) Which element is reduced in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

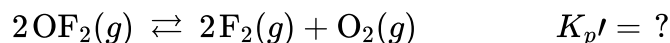
Time	P_{F_2} (atm)	P_{O_2} (atm)	P_{OF_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.79	0.90	0.21
t_2	0.62	0.81	0.38
t_3	0.50	0.75	0.50
t_4	0.41	0.71	0.59
t_5	0.37	0.69	0.63
t_6	0.35	0.68	0.65
t_7	0.34	0.67	0.66
t_8	0.33	0.67	0.67
t_9	0.33	0.67	0.67

- (c) At time t_2 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds fewer molecules of which gas, $\text{F}_2(g)$ or $\text{O}_2(g)$? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is increased after time t_9 , the mole fraction of OF_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{F}_2(g)$ or $\text{OF}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of the number of moles of O_2 .

$$3.8 \text{ g F}_2 \times \frac{1 \text{ mol F}_2}{38 \text{ g F}_2} = 0.10 \text{ mol F}_2$$

$$0.10 \text{ mol F}_2 \times \frac{1 \text{ mol O}_2}{2 \text{ mol F}_2} = 0.050 \text{ mol O}_2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response both the correct answer and a valid justification equivalent to the following.

· Fluorine. Its oxidation number changes from 0 to -1 .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification equivalent to one of the following.

· Greater than. The partial pressure of the product increases from time t_2 to t_3 , indicating that the forward reaction is

Chapter 10 Part 2: Le Chatelier's Principle FRQ

occurring at a greater rate than the reverse reaction.

OR

· Greater than. The partial pressures of the reactants decrease from time t_2 to t_3 , indicating that the forward reaction is occurring at a greater rate than the reverse reaction.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· F_2 . At equilibrium (t_7 , t_8 , t_9) the partial pressure of F_2 is lowest, and thus the number of molecules of F_2 is lowest.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. According to Le Châtelier's principle, the distribution of species will shift towards products when the temperature is increased in an endothermic reaction. Therefore, the mole fraction of OF_2 will increase.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of K_p .

$$\cdot K_p = \frac{(P_{OF_2})^2}{(P_{F_2})^2(P_{O_2})} = \frac{(0.67)^2}{(0.33)^2(0.67)} = 6.2$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response shows **one** of the following.

- ☐ $K_p' = \frac{(P_{F_2})^2(P_{O_2})}{(P_{OF_2})^2} = \frac{(0.33)^2(0.67)}{(0.67)^2} = 0.16$
- ☐ $K_p' = \frac{1}{K_p} = \frac{1}{6.2} = 0.16$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

$\cdot F_2(g)$ will behave more like an ideal gas. F_2 has dispersion forces only, while OF_2 has dipole-dipole attractions as well as stronger dispersion forces, due to a larger and more polarizable electron cloud. Therefore, F_2 molecules have weaker overall intermolecular forces than OF_2 .

Chapter 10 Part 2: Le Chatelier's Principle FRQ

16. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{H}_2(g)$ and $\text{Br}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



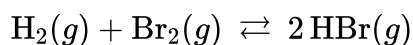
- (a) If 1.6 g of Br_2 is consumed in the reaction with excess H_2 , how many moles of HBr are produced?
- (b) Which element is reduced in this reaction? Justify your answer in terms of the oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

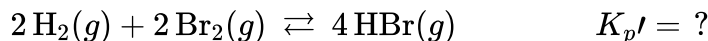
Time	P_{H_2} (atm)	P_{Br_2} (atm)	P_{HBr} (atm)
t_0	1.0	1.0	0.0
t_1	0.78	0.78	0.44
t_2	0.62	0.62	0.76
t_3	0.51	0.51	0.98
t_4	0.45	0.45	1.10
t_5	0.42	0.42	1.17
t_6	0.41	0.41	1.19
t_7	0.40	0.40	1.20
t_8	0.40	0.40	1.20
t_9	0.40	0.40	1.20

- (c) At time t_9 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.
- (d) At equilibrium, the container holds the most molecules of which gas, H_2 , Br_2 , or HBr ? Explain your answer.
- (e) A student hypothesizes that if the temperature of the container is decreased after time t_9 , the mole fraction of HBr in the container will decrease. Do you agree or disagree with the student's hypothesis? Justify your answer.
- (f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{Br}_2(g)$ or $\text{H}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of the number of moles of HBr.

$$1.6 \text{ g Br}_2 \times \frac{1 \text{ mol Br}_2}{160 \text{ g Br}_2} = 0.010 \text{ mol Br}_2$$

$$0.010 \text{ mol Br}_2 \times \frac{2 \text{ mol HBr}}{1 \text{ mol Br}_2} = 0.020 \text{ mol HBr}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Bromine. Its oxidation number changes from 0 to -1 .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Equal to. The partial pressure of the product (or reactants) remains the same from time t_8 to t_9 , indicating that the

Chapter 10 Part 2: Le Chatelier's Principle FRQ

reaction is at equilibrium.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· HBr. At equilibrium (t_7 , t_8 , t_9) the partial pressure of HBr is greatest, and thus the number of molecules of HBr is greatest.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. According to Le Châtelier's principle, the distribution of species will shift towards products when the temperature is decreased in an exothermic reaction. Therefore, the mole fraction of HBr will increase, not decrease.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of K_p .

$$\cdot K_p = \frac{(P_{\text{HBr}})^2}{(P_{\text{H}_2})(P_{\text{Br}_2})} = \frac{(1.20)^2}{(0.40)(0.40)} = 9.0$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response shows **one** of the following.

- ☐ $K_p' = \frac{(P_{\text{HBr}})^4}{(P_{\text{H}_2})^2(P_{\text{Br}_2})^2} = \frac{(1.20)^4}{(0.40)^2(0.40)^2} = 81$
- ☐ $K_p' = (K_p)^2 = (9.0)^2 = 81$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



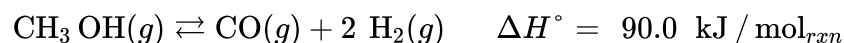
0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· $\text{H}_2(g)$ will behave more like an ideal gas. H_2 has a smaller and less polarizable electron cloud, it has weaker London dispersion forces than Br_2 . Therefore, H_2 molecules have weaker overall intermolecular forces than Br_2 .

Chapter 10 Part 2: Le Chatelier's Principle FRQ

17. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Methanol vapor decomposes to form carbon monoxide gas and hydrogen gas at high temperatures in the presence of a platinum catalyst, as represented by the balanced chemical equation given.

(a) Are the hydrogen atoms oxidized or are they reduced in the forward reaction? Justify your answer in terms of oxidation numbers.

(b) In the following box, draw the complete Lewis electron-dot diagram for the carbon monoxide molecule in which every atom obeys the octet rule. Show all bonding and nonbonding valence electrons.



(c) The values of the standard molar entropies of the compounds involved in the reaction are given in the following table.

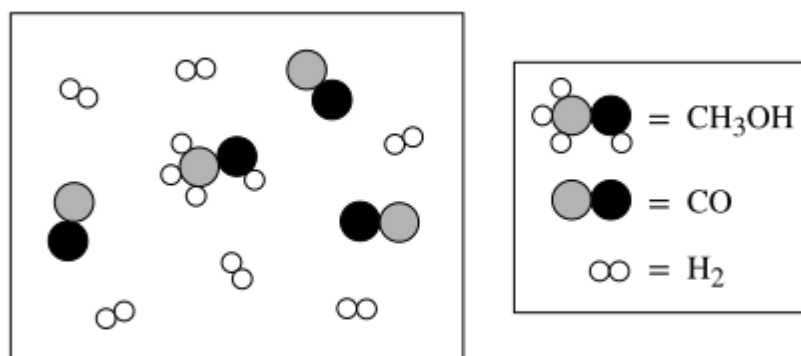
Substance	S° (J/(K·mol))
$\text{CH}_3\text{OH}(g)$	240.
$\text{CO}(g)$	198
$\text{H}_2(g)$	131

(i) Use the data in the table to calculate the value of the standard entropy change, ΔS° , in J/(K·mol_{rxn}), for the reaction.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(ii) Calculate the value of ΔG° , in $\text{kJ/mol}_{\text{rxn}}$, for the reaction at 375 K. Assume that ΔH° and ΔS° are independent of temperature.

The following particle-level diagram shows a representative sample of the equilibrium mixture represented by the equation given.



(d) Use information from the particle diagram to calculate the partial pressure of CO at equilibrium when the total pressure of the equilibrium mixture is 12.0 atm.

(e) Write the expression for the equilibrium constant, K_p , for the reaction.



The reaction system represented by the equation is allowed to achieve equilibrium at a different temperature. The following table gives the partial pressure of each species in the equilibrium mixture.

Substance	Partial Pressure at Different Temperature
$\text{CH}_3\text{OH}(g)$	2.7 atm
$\text{CO}(g)$	4.2 atm
$\text{H}_2(g)$	8.4 atm

(f) Use the information in the table to calculate the value of the equilibrium constant, K_p , at the new temperature.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(g) The volume of the container is rapidly doubled with no change in temperature. As equilibrium is re-established, does the number of moles of $\text{CH}_3\text{OH}(g)$ increase, decrease, or remain the same? Justify your answer by comparing the value of the reaction quotient, Q , with the value of the equilibrium constant, K_p .

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

- The H atoms are reduced because they change from an oxidation number of +1 to 0.

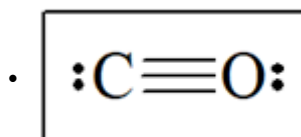
Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct diagram.



Part C(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1	2
---	---	---

The response meets **both** of the following criteria.

- ☐ The response provides the correct stoichiometry (may be implicit).
- $\Delta S_{rxn}^{\circ} = \Sigma \Delta S_{products}^{\circ} - \Sigma \Delta S_{reactants}^{\circ}$
 - $\Delta S_{rxn}^{\circ} = (\Delta S_{CO(g)}^{\circ} + 2(\Delta S_{H_2(g)}^{\circ})) - (\Delta S_{CH_3OH(g)}^{\circ})$
- ☐ The response provides the correct calculated value.
- $\Delta S_{rxn}^{\circ} = 198 + 2(131) - 240. = 220. \frac{J}{K \cdot mol_{rxn}}$

Part C(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

$$\Delta G^{\circ} = 90.0 \frac{kJ}{mol_{rxn}} - (375 K)(0.220 \frac{kJ}{K \cdot mol_{rxn}}) = +7.5 kJ/mol_{rxn}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$P_{CO} = \frac{3}{10} (12.0 \text{ atm}) = 3.6 \text{ atm}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides the correct expression.

$$\bullet K_p = \frac{(P_{\text{CO}})(P_{\text{H}_2})^2}{(P_{\text{CH}_3\text{OH}})}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$\bullet K_p = \frac{(P_{\text{CO}})(P_{\text{H}_2})^2}{(P_{\text{CH}_3\text{OH}})} = \frac{(4.2)(8.4)^2}{(2.7)} = 110$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets **both** of the following criteria.

- ☐ The response provides a correct comparison of Q and K .
- The change in volume causes the partial pressure of each species to decrease by a factor of two. Because there are more moles of gaseous products than reactants, the decrease of the numerator in Q will be larger than that in the denominator, making $Q_p < K_p$.

OR

$$Q_p = \frac{\left(\frac{P_{\text{CO}}}{2}\right)\left(\frac{P_{\text{H}_2}}{2}\right)^2}{\left(\frac{P_{\text{CH}_3\text{OH}}}{2}\right)} = \frac{K_p}{4} \approx 27 < K_p$$

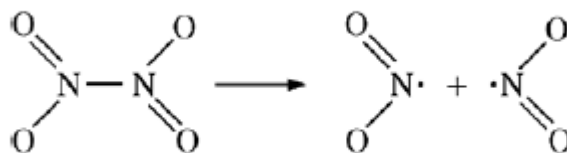
- ☐ The response provides the correct answer and a valid justification.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

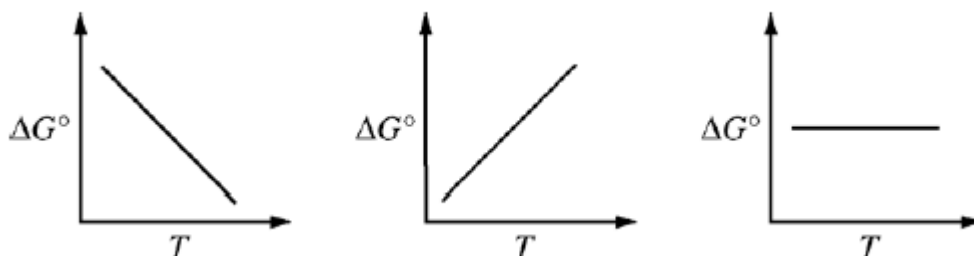
- Decrease. Given that $Q_p < K_p$, the partial pressures (moles) of the products will increase as equilibrium re-establishes, decreasing the number of moles of CH_3OH .

18. Use principles of thermodynamics to answer the following questions.

- a. The gas N_2O_4 decomposes to form the gas NO_2 according to the equation below.



- Predict the sign of ΔH° for the reaction. Justify your answer.
 - Predict the sign of ΔS° for the reaction. Justify your answer.
- b. One of the diagrams below best represents the relationship between ΔG° and temperature for the reaction given in part (a). Assume that ΔH° and ΔS° are independent of temperature.



Draw a circle around the correct graph. Explain why you chose that graph in terms of the relationship $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$.

- c. A reaction mixture of N_2O_4 and NO_2 is at equilibrium. Heat is added to the mixture while the mixture is maintained at constant pressure.
- Explain why the concentration of N_2O_4 decreases.
 - The value of K_{eq} at 25°C is 5.0×10^{-3} . Will the value of K_{eq} at 100°C be greater than, less than, or equal to this value?
- d. Using the value of K_{eq} at 25°C given in part (c)(ii), predict whether the value of ΔH° is expected to be greater than, less than, or equal to the value of $T\Delta S^\circ$. Explain.

Chapter 10 Part 2: Le Chatelier's Principle FRQ**Part A**

- i. One point is earned for the correct sign and a correct explanation.

Bonds are broken when NO_2 molecules form from N_2O_4 molecules. Energy must be absorbed to break bonds, so the reaction is endothermic and the sign of ΔH° is positive.

- ii. One point is earned for the correct sign and a correct explanation.

There are two gaseous product molecules for each gaseous reactant molecule, so the product has more entropy than the reactant. The entropy increases as the reaction proceeds, so the sign of ΔS° is positive.



0	1	2
---	---	---

Student earns 2/2 points

1. One point is earned for the correct sign and a correct explanation.

Bonds are broken when NO_2 molecules form from N_2O_4 molecules. Energy must be absorbed to break bonds, so the reaction is endothermic and the sign of ΔH° is positive.

2. One point is earned for the correct sign and a correct explanation.

There are two gaseous product molecules for each gaseous reactant molecule, so the product has more entropy than the reactant. The entropy increases as the reaction proceeds, so the sign of ΔS° is positive.

Part B

One point is earned for the correct graph selection.

One point is earned for the explanation.

The leftmost graph should be circled.

ΔS° is positive, so as T increases, $T\Delta S^\circ$ becomes a larger positive number. At higher temperatures, you are subtracting larger positive numbers from ΔH° to get ΔG° , so ΔG° decreases with increasing temperature.



0	1	2
---	---	---

Student earns 2/2 points

One point is earned for the correct graph selection.

One point is earned for the explanation.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The leftmost graph should be circled.

ΔS° is positive, so as T increases, $T\Delta S^\circ$ becomes a larger positive number. At higher temperatures, you are subtracting larger positive numbers from ΔH° to get ΔG° , so ΔG° decreases with increasing temperature.

Part C

- i. One point is earned for the correct explanation.

The reaction is endothermic. For endothermic reactions, increasing the temperature drives the reaction to the right. This increases the equilibrium concentration of NO_2 and decreases the equilibrium concentration of N_2O_4 .

- ii. One point is earned for the correct choice. (No explanation required.)

Because the reaction is endothermic, at higher temperatures the reaction goes further to the right. This means that the value of K_{eq} at 100°C will be greater than the value of K_{eq} at 25°C .



0	1	2
---	---	---

Student earns 2/2 points

1. One point is earned for the correct explanation.

The reaction is endothermic. For endothermic reactions, increasing the temperature drives the reaction to the right. This increases the equilibrium concentration of NO_2 and decreases the equilibrium concentration of N_2O_4 .

2. One point is earned for the correct choice. (No explanation required.)

Because the reaction is endothermic, at higher temperatures the reaction goes further to the right. This means that the value of K_{eq} at 100°C will be greater than the value of K_{eq} at 25°C .

Part D

One point is earned for the correct prediction.

One point is earned for the explanation.

K_{eq} at 25°C is less than 1, hence ΔG° must be positive. And in order for ΔG° to be positive, ΔH° must be greater than $T\Delta S^\circ$.



0	1	2
---	---	---

Student earns 2/2 points

One point is earned for the correct prediction.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

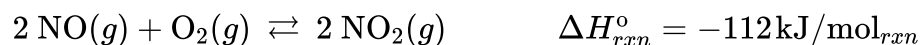
One point is earned for the explanation.

K_{eq} at 25°C is less than 1, hence ΔG° must be positive. And in order for ΔG° to be positive, ΔH° must be greater than $T\Delta S^\circ$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

19. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Equimolar amounts of $\text{NO}(g)$ and $\text{O}_2(g)$ are injected into an evacuated, rigid container, where they react according to the equation below.



(a) If 3.2 g of $\text{O}_2(g)$ is consumed in the reaction with excess $\text{NO}(g)$, how many moles of $\text{NO}_2(g)$ are produced?

(b) Which element is oxidized in this reaction? Justify your answer in terms of oxidation numbers.

The partial pressures of the gases in the container are monitored at constant temperature and recorded in the table below.

Time	P_{NO} (atm)	P_{O_2} (atm)	P_{NO_2} (atm)
t_0	1.0	1.0	0.0
t_1	0.80	0.90	0.20
t_2	0.69	0.84	0.31
t_3	0.62	0.81	0.38
t_4	0.58	0.79	0.42
t_5	0.55	0.77	0.45
t_6	0.52	0.76	0.48
t_7	0.50	0.75	0.50
t_8	0.50	0.75	0.50
t_9	0.50	0.75	0.50

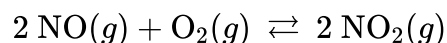
(c) At time t_3 , is the rate of the forward reaction greater than, less than, or equal to the rate of the reverse reaction? Justify your choice.

(d) At equilibrium, the container holds the most molecules of which gas, NO , O_2 , or NO_2 ? Explain your answer.

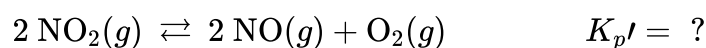
Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) A student hypothesizes that if the temperature of the container is increased after time t_9 , the mole fraction of O_2 in the container will increase. Do you agree or disagree with the student's hypothesis? Justify your answer.

(f) Using the data in the table, determine the value of the equilibrium constant, K_p , for the reaction represented by the equation below.



(g) Using the value determined in part (f), determine the value of K_p' for the equation shown below.



(h) Which gas, $\text{NO}_2(g)$ or $\text{O}_2(g)$, will behave most like an ideal gas at low temperature? Justify your choice.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of the number of moles of NO_2 .

$$3.2 \text{ g O}_2 \times \frac{1 \text{ mol O}_2}{32 \text{ g O}_2} = 0.10 \text{ mol O}_2$$

$$0.10 \text{ mol O}_2 \times \frac{2 \text{ mol NO}_2}{1 \text{ mol O}_2} = 0.20 \text{ mol NO}_2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the element that is oxidized and a valid justification equivalent to the following.

· Nitrogen. Its oxidation number changes from +2 to +4.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides both the correct answer and a valid justification equivalent to one of the following.

· Greater than. The partial pressure of the product increases from time t_3 to t_4 , indicating that the forward reaction is occurring at a greater rate.

OR

· Greater than. The partial pressures of the reactants continue to decrease from time t_3 to time t_4 , indicating that the forward reaction is occurring at a greater rate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and an explanation equivalent to the following.

· O_2 . At equilibrium (t_7 , t_8 , t_9) the partial pressure of O_2 is greatest, and thus the number of molecules of O_2 is greatest.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a valid justification equivalent to one of the following.

· Agree. According to Le Châtelier's principle, the distribution of species will shift towards reactants when the temperature is increased in an exothermic reaction. Therefore, the mole fraction of O_2 will increase.

OR

· Disagree. Increasing the temperature of the constant-volume system would increase the pressure in the container. According to Le Châtelier's principle, the distribution of species will shift towards the side with the fewer number of moles. Therefore, the mole fraction of O_2 will decrease.

Part F

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct calculation of K_p .

$$K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{NO}})^2(P_{\text{O}_2})} = \frac{(0.50)^2}{(0.50)^2(0.75)} = 1.3$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows **one** of the following.

- ☐ $K_p' = \frac{(P_{\text{NO}})^2(P_{\text{O}_2})}{(P_{\text{NO}_2})^2} = \frac{(0.50)^2(0.75)}{(0.50)^2} = 0.75$
- ☐ $K_p' = \frac{1}{K_p} = \frac{1}{1.3} = 0.75$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



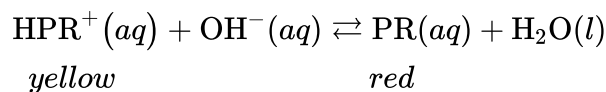
0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

$\text{O}_2(g)$ will behave more like an ideal gas because O_2 has dispersion forces only, while NO_2 has dipole-dipole attractions as well as stronger dispersion forces due to a larger and more polarizable electron cloud. Therefore, $\text{O}_2(g)$ molecules have weaker overall intermolecular forces than $\text{NO}_2(g)$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

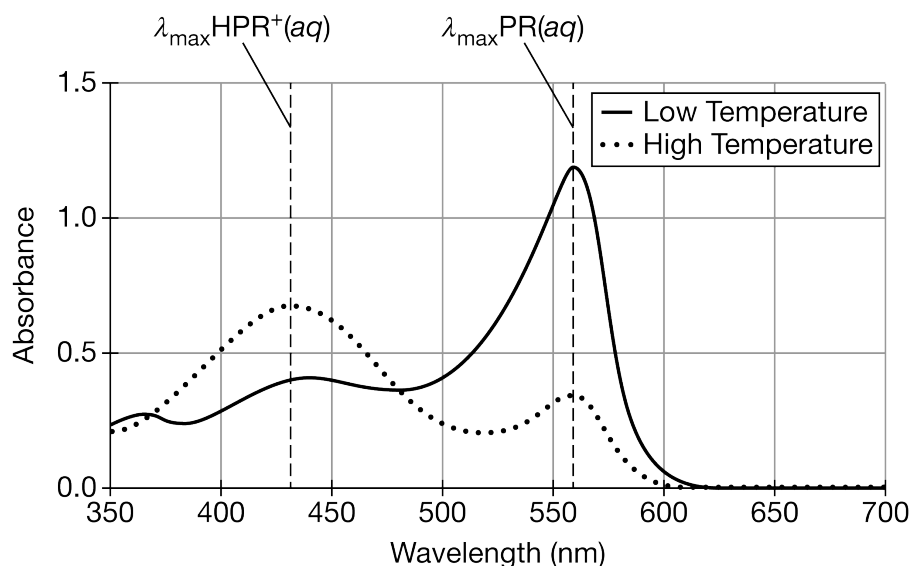
20. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The acidified form of phenol red, $\text{HPR}^+(aq)$, is yellow with a strong absorption peak at a wavelength of 435 nm. It reacts with hydroxide ions according to the equation above, producing water and phenol red (PR), which has a strong absorption peak at a wavelength of 555 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

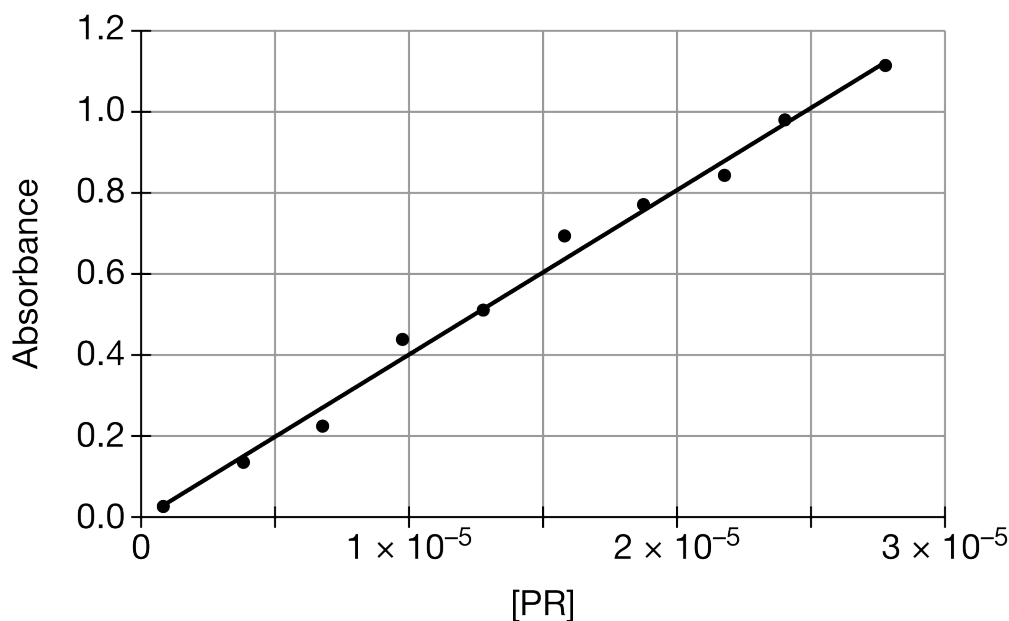
The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 365 nm or the one at 435 nm, results from the absorption of photons with the greater energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[\text{PR}] = 1.5 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of **PR** of unknown concentration in order to determine the $[\text{PR}]$ in the solution using the calibration plot above. The cuvette is rinsed with water, then with a small amount of the solution, which is discarded. Finally, the cuvette is filled with the mixture for analysis. Explain why rinsing the cuvette with a small amount of solution is important to ensure an accurate absorbance measurement.

Over time, **PR** converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the **PR** solution at 555 nm in a 2.00 M KOCCl solution. The temperature and $[\text{OCl}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (hr)	Absorbance
0	0.40
0.40	0.32
0.80	0.25
1.20	0.20
1.60	0.16
2.00	0.13
2.40	0.10

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 0.500 M . The observed half-life of the reaction is longer than for the first trial when $[\text{OCl}^-]$ was 2.00 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction increases when $[\text{OCl}^-]$ is changed from 2.00 M to 0.500 M .

(i) Is the activation energy for the reaction when $[\text{OCl}^-]$ is 2.00 M greater than, the same as, or less than the activation energy when $[\text{OCl}^-]$ is 0.500 M ? Justify your answer.

In order to prepare the original 2.00 M solution of $\text{KOC}\text{l}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 90.6 g of $\text{KOC}\text{l}(s)$.

Step 2: Add 1.00 L of distilled water to a 2.000 L volumetric flask.

Step 3: Add the $\text{KOC}\text{l}(s)$ to the flask and swirl to dissolve the $\text{KOC}\text{l}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) The chemical quantities used in the procedure do NOT produce $2.00\text{ M KOC}\text{l}(aq)$. Describe specifically how they should be changed in order to produce $2.00\text{ M KOC}\text{l}(aq)$.

(k) Another student is preparing a $2.00\text{ M KOC}\text{l}$ solution using the correct chemical quantities but notices that some solid remains at the bottom of the volumetric flask. Describe an action that the student should take to ensure that the remaining solid dissolves.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KCl}(s)$ in $1000. \text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below

Solvent	X	Y
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	1.8	2.2

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{PR}]}{[\text{HPR}^+][\text{OH}^-]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

$\cdot 365 \text{ nm}$. Photons with a wavelength of 365 nm have a higher frequency than those with a wavelength of 435 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the higher frequency (shorter wavelength) photons have the greater energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

· Exothermic. The diagram shows that the absorbance, and therefore the concentration of HPR^+ is greater at high temperature. The increased concentration of reactants at elevated temperature is consistent with an exothermic reaction, according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly explains why the procedure is important for an accurate measurement.

· Accurate absorbance readings depend on measuring solutions of the correct concentration. Rinsing the cuvette with some of the solution to be analyzed ensures that there is no dilution of the solution by residual water inside the cuvette, which would lead to lower measured absorbance.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

- ☐ The response indicates a value of 1.20 for the half-life
- ☐ The response has units of “hr” or “hours”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first-order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first-order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OCl^- decreases, the collisions between reactant particles occur less frequently per unit of time. The lower frequency of collisions leads to a decreased rate of product formation and correspondingly a longer half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· Activation energy stays the same. Activation energy does not change with concentration. (Only a catalyst or different reaction pathway could alter the activation energy.)

Part J

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid correction to the chemical quantities.

· To make 2.00 M KOCℓ(aq), the student needs to dissolve 181 g (2.00 mol) of KOCℓ in 1.00 L of solution.

OR

· To make 2.00 M KOCℓ(aq), the student needs to dissolve 362 g (4.00 mol) of KOCℓ in 2.00 L of solution.

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid experimental procedure.

· Gently swirl or stir the solution to dissolve the remaining solid in the flask.

OR

· Stopper the flask and invert several times until the solid dissolves.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



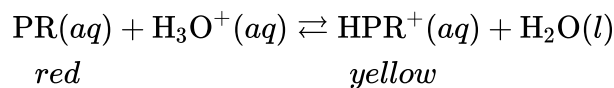
0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The temperature change would be greater in solvent X than in solvent Y. Since solvent X has the smaller heat capacity, the same energy transferred to an equal mass of it will result in a greater temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

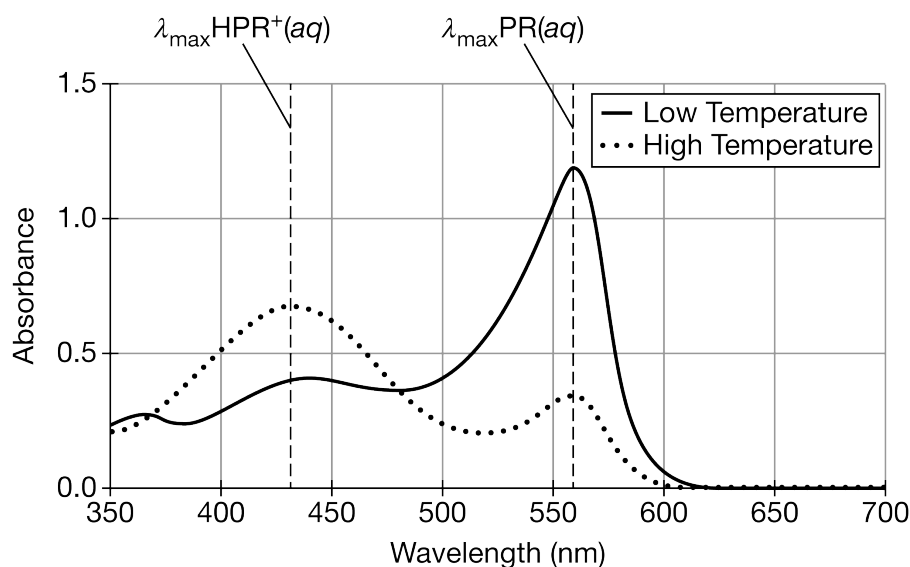
21. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



Phenol red, $\text{PR}(aq)$, is a red compound that has a strong absorption peak at a wavelength of 555 nm. It reacts with hydronium ions according to the equation above, producing water and the acidified form of phenol red, $\text{HPR}^+(aq)$, which is yellow and has a strong absorption peak at a wavelength of 435 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

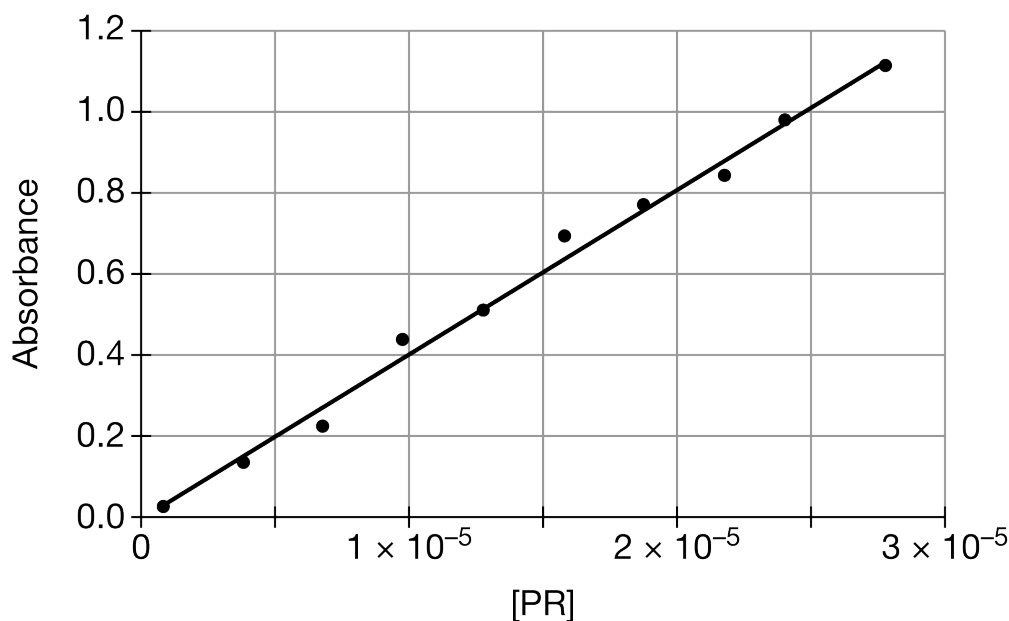
The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 435 nm or the one at 555 nm, results from the absorption of photons with the greater energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at wavelength 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[\text{PR}] = 2.0 \times 10^{-5} \text{ M}$?

(e) A student measures the absorbance of a solution of PR of unknown concentration in order to determine the $[\text{PR}]$ in the solution using the calibration plot above. The outside of the cuvette is wiped clean before placing it in the spectrophotometer. Explain why this step is important to ensure an accurate absorbance measurement.

Over time, PR converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the PR solution at 555 nm in a 0.50 M KOCℓ solution. The temperature and $[\text{OCl}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (min)	Absorbance
0	0.80
6	0.63
12	0.50
18	0.40
24	0.32
30	0.25
36	0.20

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 1.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OCl}^-]$ was 0.50 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OCl}^-]$ is changed from 0.50 M to 1.00 M .

(i) When $[\text{OCl}^-]$ is changed from 0.50 M to 1.00 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the $1.00\text{ M KOCl}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 45.3 g of $\text{KOCl}(s)$.

Step 2: Add the $\text{KOCl}(s)$ to a 1000.0 mL volumetric flask that is half-filled with distilled water and swirl to dissolve.

Step 3: Add enough distilled water to bring the volume up to the line on the flask.

(j) The chemical quantities used in the procedure do NOT produce $1.00\text{ M KOCl}(aq)$. Describe specifically how they should be changed in order to produce $1.00\text{ M KOCl}(aq)$.

(k) Another student is preparing a KOCl solution using the correct chemical quantities but notices that some undissolved solid remains at the bottom of the volumetric flask. Under these circumstances, is the concentration of $\text{KOCl}(aq)$ in the solution in the flask greater than, less than, or equal to 1.00 M ? Explain your answer.

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KOCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KOCl}(s)$ in $1000.\text{ g}$ each of two different

Chapter 10 Part 2: Le Chatelier's Principle FRQ

solvents, water and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in water be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of water and solvent Y are given in the table below.

Solvent	Water	Y
Specific Heat Capacity (J/(g · °C))	4.2	2.2

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{HPR}^+]}{[\text{PR}][\text{H}_3\text{O}^+]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

· 435 nm. Photons with a wavelength of 435 nm have a higher frequency than those with a wavelength of 555 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the higher frequency (shorter wavelength) photons have the greater energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct choice and a valid justification equivalent to the following.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

· Endothermic. The diagram shows that the absorbance, and therefore the concentration, of HPR^+ is greater at high temperature. The increased concentration of products at elevated temperature is consistent with an endothermic reaction, according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· 0.80

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly describes the effect of the error.

· Material on the outside of the cuvette (such as a smudge, fingerprint, or water droplet) could scatter or absorb some of the incoming light, making the measured absorbance greater than it should be. Because absorbance and concentration are directly proportional, the greater absorbance would be misinterpreted as a concentration of PR that is higher than the actual concentration.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets both of the following criteria.

- ☐ The response indicates a value of 18 for the half-life
- ☐ The response has units of “min” or “minutes”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OCl^- increases, the collisions between reactant particles occur more frequently per unit of time. The greater frequency of collisions leads to an increased rate of product formation and correspondingly a shorter half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· The magnitude of ΔH_{rxn}° will remain the same. ΔH_{rxn}° is an intensive property, depending on both the enthalpy change and the number of moles reacted. Regardless of the number of moles that react, the ratio of the enthalpy change per mole will always be the same.

OR

· The magnitude of ΔH_{rxn}° will remain the same. Although the $[\text{OCl}^-]$ is increased, the number of moles of each reactant that is consumed is the same, as OCl^- is not the limiting reactant. Hence, the magnitude of the enthalpy change per mole would be the same.

Part J

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid correction to the chemical quantities.

· To make 1.00 *M* KOCℓ(*aq*), the student needs to dissolve 90.6 g (1.00 mol) of KOCℓ in 1.00 L of solution.

OR

· To make 1.00 *M* KOCℓ(*aq*), the student needs to dissolve 45.3 g (0.500 mol) of KOCℓ in 0.500 L of solution.

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· Less than 1.00 *M*. There are fewer moles of KOCℓ dissolved in the solution than was intended.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



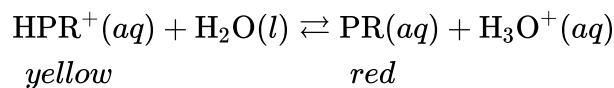
0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The temperature change would be less in water than in solvent Y. Since water has the greater heat capacity, the same energy transferred to an equal mass of it will result in a smaller temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

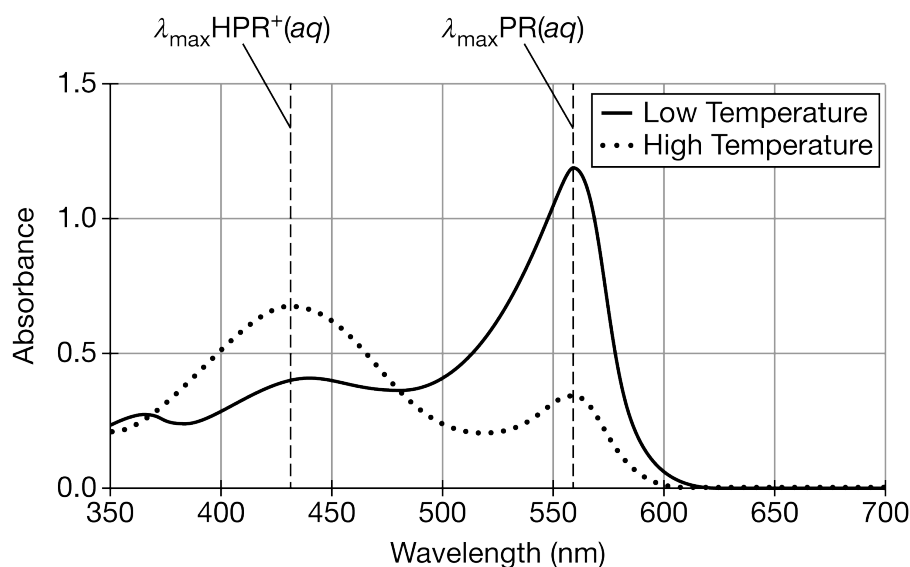
22. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The acidified form of phenol red, $\text{HPR}^+(aq)$, is yellow with a strong absorption peak at a wavelength of 435 nm. It reacts with water according to the equation above, producing hydronium ions and phenol red (PR), which has a strong absorption peak at a wavelength of 555 nm.

a) Write the K_{eq} expression for the equilibrium reaction above.

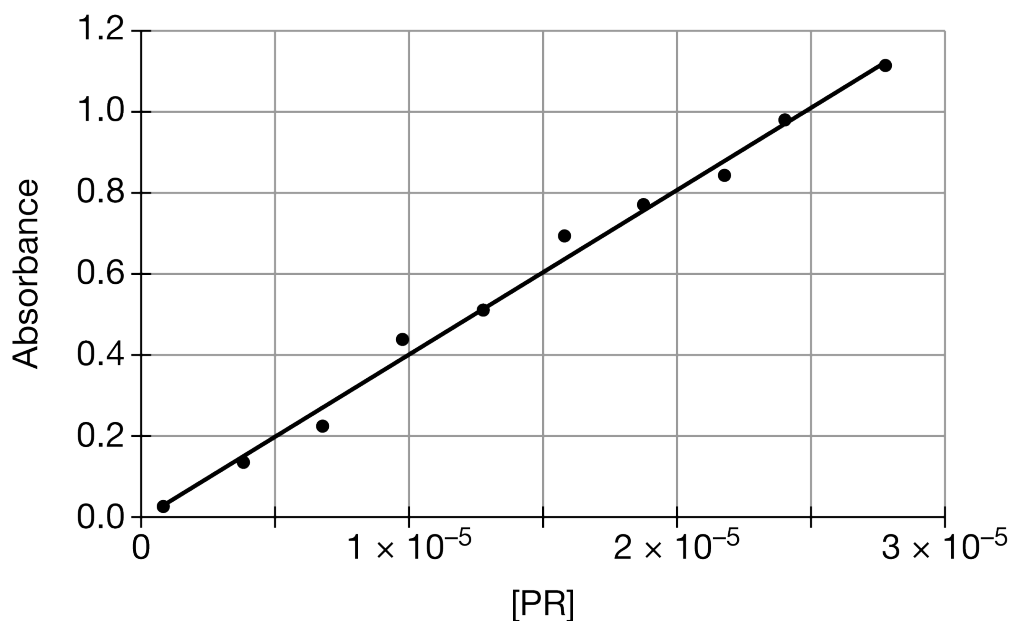
The absorption spectra of an equilibrium mixture of $\text{PR}(aq)$ and $\text{HPR}^+(aq)$, collected at two different temperatures, are shown in the following diagram.



- (b) Which peak, the one at 435 nm or the one at 555 nm, results from the absorption of photons with the lower energy? Justify your answer.
- (c) Based on the diagram above, is the forward reaction endothermic or exothermic? Justify your answer in terms of Le Châtelier's principle.

A calibration plot of the absorbance at 555 nm versus the concentration of $\text{PR}(aq)$ is shown in the following graph.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Based on the best-fit line in the graph, what is the approximate absorbance value for a solution in which $[\text{PR}] = 0.5 \times 10^{-5} M$?

(e) A student measures the absorbance of a solution of **PR** of unknown concentration in order to determine the $[\text{PR}]$ in the solution using the calibration plot above. The cuvette used for making this measurement has the same path length as the cuvette used in making the calibration plot. Explain why this step is important to ensure an accurate absorbance measurement.

Over time, **PR** converts to a colorless molecule in the presence of OCl^- ions. To study the rate of the reaction, the student recorded the absorbance of the **PR** solution at 555 nm in a 1.00 M KOCCl solution. The temperature and $[\text{OCl}^-]$ remain constant throughout the reaction. The results are summarized in the table below.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Time (s)	Absorbance
0	0.80
120	0.63
240	0.50
360	0.40
480	0.32
600	0.25
720	0.20

(f) Determine the half-life of the reaction. Include units in your answer.

(g) Explain how the absorbance and time data are consistent with a process that is first order with respect to PR.

A second trial is conducted at the same constant temperature, but with $[\text{OCl}^-]$ changed to 2.00 M . The observed half-life of the reaction is shorter than for the first trial when $[\text{OCl}^-]$ was 1.00 M .

(h) Explain, in term of particle collisions, why the half-life of the reaction decreases when $[\text{OCl}^-]$ is changed from 1.00 M to 2.00 M .

(i) When $[\text{OCl}^-]$ is changed from 1.00 M to 2.00 M , does the magnitude of $\Delta H^\circ_{\text{rxn}}$, the enthalpy change of the reaction, increase, decrease, or stay the same? Justify your answer.

In order to prepare the 2.00 M solution of $\text{KOC}\text{l}(aq)$, a student follows a procedure with the following steps.

Step 1: Measure out 18.11 g of $\text{KOC}\text{l}(s)$.

Step 2: Add the $\text{KOC}\text{l}(s)$ to a clean, dry 100.0 mL volumetric flask.

Step 3: Add about 50 mL of distilled water to the flask and swirl to dissolve the $\text{KOC}\text{l}(s)$.

Step 4: Add enough distilled water to bring the mixture to the line on the volumetric flask, inverting several times to mix.

(j) Will the chemical quantities in the procedure produce a solution that is $2.00\text{ M KOC}\text{l}(aq)$? Explain your reasoning.

(k) Before adding the $\text{KOC}\text{l}(s)$ to the volumetric flask, the student first rinsed the volumetric flask with distilled water, leaving some droplets of distilled water on the inside of the flask. As a result of this step, will the concentration of $\text{KOC}\text{l}(aq)$ be greater than, less than, or equal to 2.00 M ? Explain your answer.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(l) When making the solution above, the student notices that the temperature increases when dissolving $\text{KCl}(s)$ in water. In a separate experiment, the student dissolves 1.0 mol of $\text{KCl}(s)$ in $1000. \text{ g}$ each of two different solvents, X and Y. Assuming the enthalpy change is the same regardless of solvent, would the temperature change in solvent X be greater than, less than, or equal to the temperature change in solvent Y? Justify your answer. The specific heat capacities of solvent X and solvent Y are given in the table below.

Solvent	X	Y
Specific Heat Capacity ($\text{J}/(\text{g} \cdot ^\circ\text{C})$)	2.2	1.8

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct K_{eq} expression.

$$K_{eq} = \frac{[\text{PR}][\text{H}_3\text{O}^+]}{[\text{HPR}^+]}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

555 nm . Photons with a wavelength of 555 nm have a lower frequency than those with a wavelength of 435 nm because frequency and wavelength are inversely proportional ($\nu = \frac{c}{\lambda}$). Since frequency and energy are directly proportional ($E = h\nu$), the higher frequency (shorter wavelength) photons have the greater energy.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· Exothermic. The diagram shows that the absorbance, and therefore the concentration, of HPR^+ is greater at high temperature. The increased concentration of reactants at elevated temperature is consistent with an exothermic reaction, according to Le Châtelier's principle.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct absorbance.

· 0.20

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response correctly explains why the procedure is important for an accurate measurement.

· According to Beer's law, absorbance and the path length of the cuvette are directly proportional. Thus, identical samples will have different absorbance values if the path length of the cuvette is not the same. Keeping the cuvette path length constant in every experiment ensures that differences in absorbance readings arise only from differences in sample concentration.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The response meets both of the following criteria.

- ☐ The response indicates a value of 360 for the half-life
- ☐ The response has units of “s” or “seconds”

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response meets one of the following criteria.

- ☐ The response indicates that the half-life of the reaction remains constant throughout the reaction, and that this characteristic is unique to first order reactions.
- ☐ The response indicates that the half-life of this reaction is independent of concentration, and that this characteristic is unique to first order reactions.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a valid explanation equivalent to the following.

· When the concentration of OCl^- increases, the collisions between reactant particles occur more frequently per unit of time. The greater frequency of collisions leads to an increased rate of product formation and correspondingly a shorter half-life of the reaction.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· The magnitude of ΔH_{rxn}° will remain the same. ΔH_{rxn}° is an intensive property, depending on both the enthalpy change and the number of moles reacted. Regardless of the number of moles that react, the ratio of the enthalpy change

Chapter 10 Part 2: Le Chatelier's Principle FRQ

per mole will always be the same.

OR

· The magnitude of ΔH_{rxn}° will remain the same. Although the $[\text{OCl}^-]$ is increased, the number of moles of each reactant that is consumed is the same, as OCl^- is not the limiting reactant. Hence, the magnitude of the enthalpy change per mole would be the same.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice with a valid justification equivalent to the following.

· Yes. The student has dissolved 18.11 g (0.2000 mol) KOCl in 100.0 mL (0.1000 L) of solution, resulting in a concentration of 2.00 M .

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· The concentration of the KOCl solution will be equal to 2.00 M . The droplets of distilled water simply mix with the rest of the distilled water added in step 3. As long as the volumetric flask is filled to the correct mark in step 4, the concentration of the solution will be 2.00 M .

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

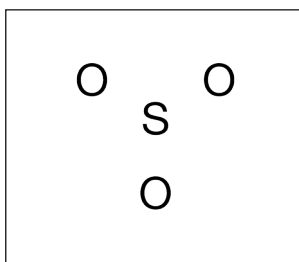
· The temperature change would be less in solvent X than in solvent Y. Since solvent X has the greater heat capacity, the same energy transferred to an equal mass of it will result in a smaller temperature change, consistent with the relationship $q = mc\Delta T$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

23. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

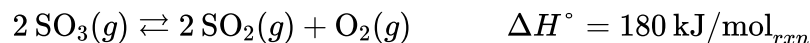
Sulfur trioxide, SO_3 , is commercially produced on a massive scale for use in manufacturing. It is also produced as a by-product of burning fossil fuels and is an air pollutant involved in the formation of acid rain.

(a) In the box below, complete one of the valid Lewis electron-dot diagrams for the SO_3 molecule. Be sure to include all valence electrons, including lone pairs.



(b) Predict the shape of the SO_3 molecule based on the diagram you completed above.

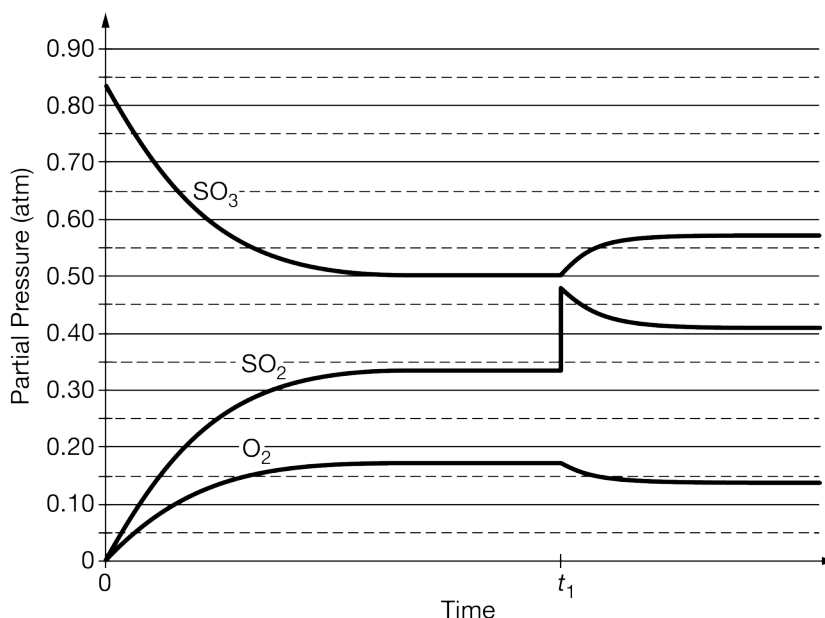
Under some conditions, $\text{SO}_3(g)$ decomposes to produce $\text{SO}_2(g)$ and $\text{O}_2(g)$, as represented by the equation below.



(c) Under what temperature conditions will the reaction be thermodynamically favored: high temperatures, low temperatures, all temperatures, or no temperatures? Justify your answer in terms of the change in enthalpy and the change in entropy during the reaction.

A chemist fills a rigid vessel with $\text{SO}_3(g)$ at a certain temperature until the pressure of $\text{SO}_3(g)$ in the container is 0.83 atm. The $\text{SO}_3(g)$ decomposes as the partial pressures of $\text{SO}_3(g)$, $\text{SO}_2(g)$, and $\text{O}_2(g)$ in the container are monitored over time, as shown in the following graph. A change is made to the system at time t_1 .

Chapter 10 Part 2: Le Chatelier's Principle FRQ



(d) Answer the following based on the graph.

(i) Draw an **X** on the time axis to show when the system first reached equilibrium.

(ii) Describe the change that was made to the system at time t_1 .

(iii) After the change was made at time t_1 , the partial pressure of $\text{SO}_3(g)$ increased while the partial pressure of $\text{O}_2(g)$ decreased. Explain this observation.

(e) Use the information above to answer the following.

(i) Write the expression for the equilibrium constant, K_p , for the reaction.

(ii) Determine the value of K_p for the reaction.

Part A

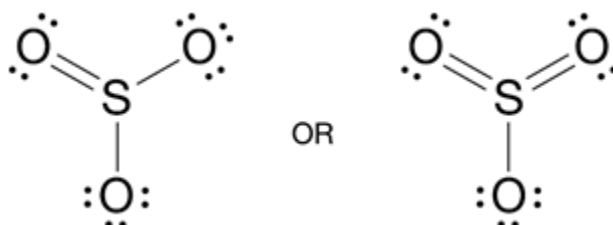
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives one of the following Lewis diagrams or an equivalent resonance form.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response gives a shape that is consistent with the answer to part (a). (The shape consistent with the answer shown above is trigonal planar.)

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response meets both of the following criteria.

- ☐ The response contains the following explanation.
The reaction will be thermodynamically favored when ΔG° is negative ($\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$). ΔH° for this reaction is positive, so at low temperatures (when the value of $T\Delta S^\circ$ is small), ΔG° is positive and the reaction is not thermodynamically favored.
- ☐ The response contains the following explanation.
Because ΔS° for this reaction is positive, high-temperature conditions give a large positive value for $T\Delta S^\circ$. This will result in a negative ΔG° , and the reaction will be thermodynamically favored.

Part D(i)

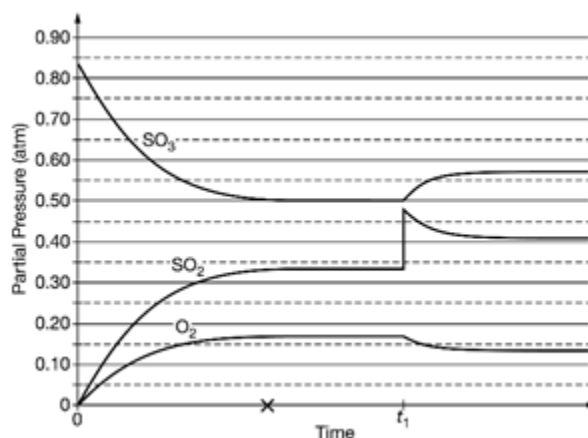
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The graph shows an X on the time axis approximately halfway between 0 and t_1 .

Chapter 10 Part 2: Le Chatelier's Principle FRQ



Part D(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response indicates that additional $\text{SO}_2(g)$ was injected into the reaction vessel.

Part D(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response indicates one of the following.

- The increase in the partial pressure of SO_2 perturbed the equilibrium, which responded by consuming O_2 and producing more SO_3 as new equilibrium concentrations were established.
- At equilibrium (before t_1), the reaction quotient Q is equal to K . Additional $\text{SO}_2(g)$ increases P_{SO_2} , increasing the value of Q such that $Q > K$. The reaction proceeds in reverse to reestablish equilibrium, consuming O_2 and producing more SO_3 .

Part E(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response includes the following expression.

$$K_p = \frac{(P_{\text{SO}_2})^2(P_{\text{O}_2})}{(P_{\text{SO}_3})^2}$$

Part E(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. If an incorrect expression from part (e)(i) is used correctly in part (e)(ii), full credit can be earned in part (e)(ii).



0	1	2
---	---	---

The response includes a calculation similar to one of the following based on appropriate values from the graph.

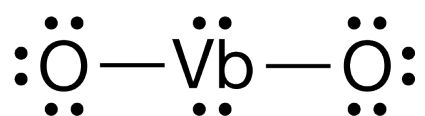
- ☐ $K_p = \frac{(0.33)^2(0.17)}{(0.50)^2} = 0.074$
- ☐ $K_p = \frac{(0.41)^2(0.14)}{(0.57)^2} = 0.072$

Chapter 10 Part 2: Le Chatelier's Principle FRQ

24. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

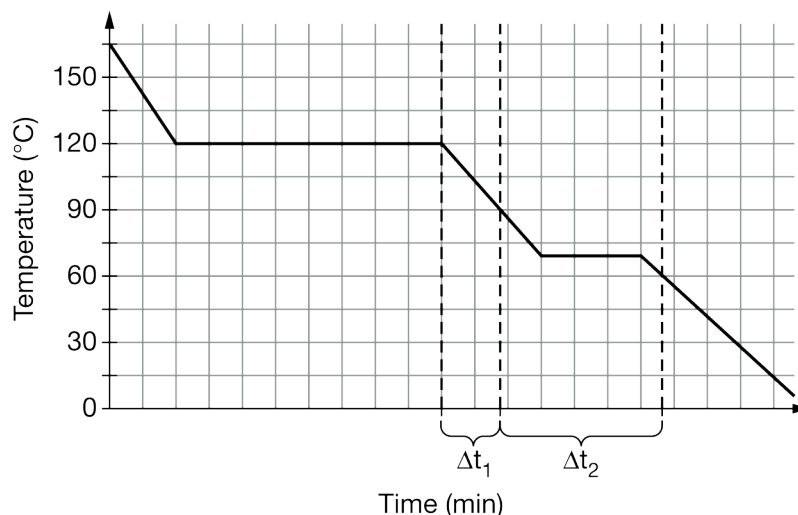
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO_2 , an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO_2 .

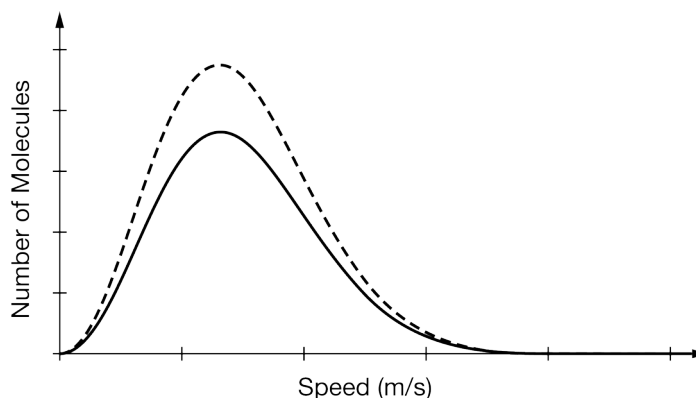
A student investigates the physical properties of VbO_2 and finds the following graph where thermal energy is removed at a constant rate from a sample of VbO_2 , which is initially a gas at 165°C . The pressure is held constant at 1.00 atm throughout the cooling process.



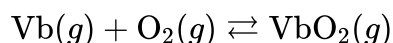
- (b) The student determines that the specific heat of $\text{VbO}_2(l)$ is $2.0 \text{ J}/(\text{g} \cdot ^\circ\text{C})$. Determine the amount of thermal energy, in J, that must be removed from a 35 g sample of $\text{VbO}_2(l)$ to reduce the temperature of the sample from $110.^\circ\text{C}$ to $80.^\circ\text{C}$.
- (c) The two intervals marked on the graph correspond to the time required for a temperature decrease of $30.^\circ\text{C}$. Given that the sample is cooled at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid SO_2 ($\Delta H_{\text{vap}}^\circ = 25 \text{ kJ/mol}$) are stronger than those in liquid VbO_2 ($\Delta H_{\text{vap}}^\circ = 32 \text{ kJ/mol}$). Do you agree or disagree with the student's claim? Justify your answer in terms of the relative values of $\Delta H_{\text{vap}}^\circ$.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO_2 particles at 165°C . The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 135°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 6.5 atm , and the initial pressure of $\text{O}_2(g)$ is 4.0 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 1.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a larger volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{Vb}(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	3.0×10^{-4}

Chapter 10 Part 2: Le Chatelier's Principle FRQ

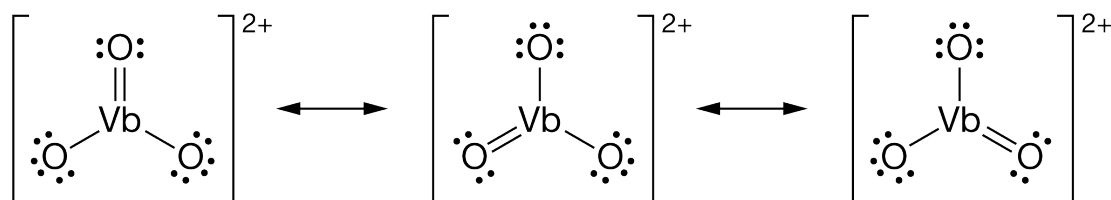
- (i) Based on the data, determine the order of the reaction with respect to $O_2(g)$. Justify your answer.
- (j) The reaction is known to be first order with respect to $Vb(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms ions with formula VbO_3^{2+} . The following Lewis diagrams represent the VbO_3^{2+} ion.



- (l) A student claims that VbO_3^{2+} contains two $Vb - O$ bonds of equal strength and one $Vb - O$ bond that is stronger than the other two. Do you agree or disagree? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct hybridization.

sp^3

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculation of thermal energy.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$q = mc\Delta T = (35 \text{ g})(2.0 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}})(80.^\circ\text{C} - 110.^\circ\text{C}) = -2100 \text{ J}.$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation equivalent to the following.

The substance undergoes a phase change (liquid to solid) during Δt_2 but not during Δt_1 . The phase change requires the removal of additional energy without changing the temperature of the substance, so the time interval for the sample to decrease by 30°C is longer in Δt_2 .

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

Disagree. The lower heat of vaporization of SO_2 (25 kJ/mol vs. 32 kJ/mol) indicates that less energy is required for the phase change from liquid to vapor. This difference arises from weaker total intermolecular attractions in SO_2 than in VbO_2 .

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation of the error equivalent to one of the following.

The maximum of the dashed line should be to the left of the maximum of the solid line. According to kinetic molecular theory, the average kinetic energy of the molecules decreases when the temperature decreases.

OR

The student incorrectly indicated a greater number of total molecules in the sample. The area under the dashed line curve should be identical to that under the solid line curve.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

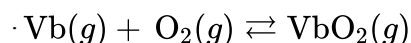
Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct partial pressures.



	Vb(g)	O ₂ (g)	VbO ₂ (g)
Initial Partial Pressure (atm)	6.5	4.0	0.0
Change in Pressure (atm)	−1.5	−1.5	+1.5
Final Partial Pressure (atm)	5.0	2.5	1.5

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct calculation of K_p consistent with the pressures in part (f).

$$K_p = \frac{P_{\text{VbO}_2}}{(P_{\text{Vb}})(P_{\text{O}_2})} = \frac{1.5}{(5.0)(2.5)} = 0.12$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Greater than. When the volume of the system increases, $Q > K$ and the distribution of species shifts towards reactants.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

OR

· Greater than. According to Le Châtelier's principle, increasing the volume of a gaseous equilibrium system will shift the distribution of species towards the side with the greatest number of gas particles. In this case, the shift favors the reactants.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct order and a valid justification equivalent to the following.

· Second order. Comparing trials 2 and 3, when the initial concentration of O_2 is halved, the initial rate is quartered.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct rate law consistent with part (i).

· $\text{Rate} = k [\text{Vb}] [\text{O}_2]^2$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a justification, consistent with part (j), equivalent to the following.

· Yes. There is a termolecular collision of one Vb and two O_2 particles, which corresponds to the respective order of each reactant in the rate law expression.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0

1

The response provides both a correct answer and a valid justification equivalent to the following.

· Disagree. The actual arrangement of electrons in VbO_3^{2+} is a blend of the three equivalent resonance structures. Thus, all three $\text{Vb} - \text{O}$ bonds are equal in strength.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

25. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

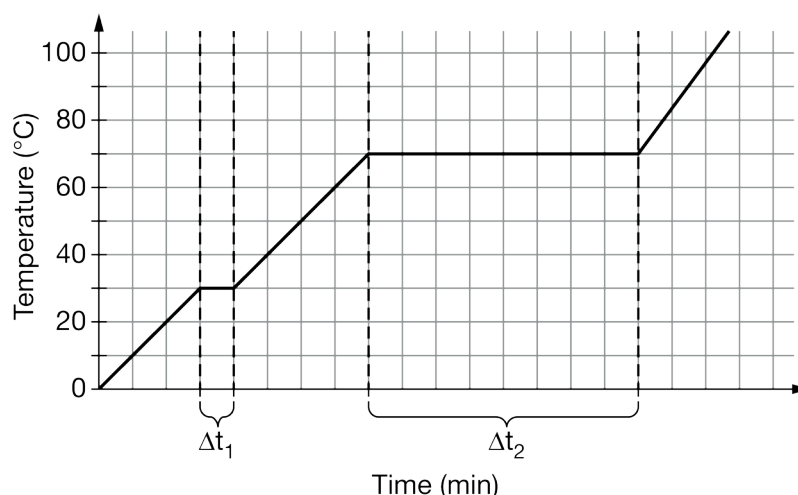
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO₂, an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO₂.

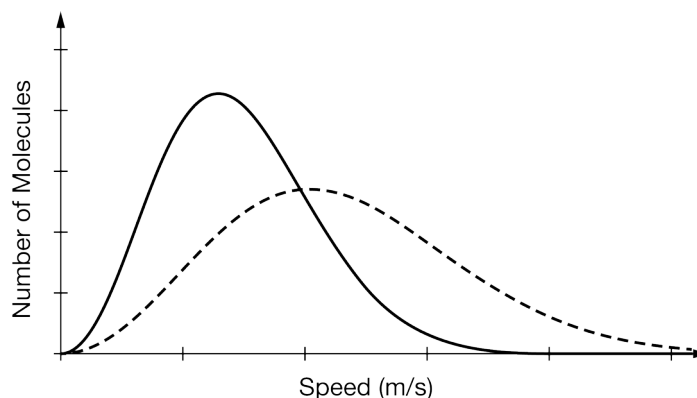
A student investigates the physical properties of VbO₂ and finds the following graph where thermal energy is added at a constant rate to a sample of VbO₂, which is initially a solid at 0 °C. The pressure is held constant at 1.00 atm throughout the heating process.



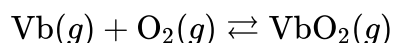
- (b) The student determines that the specific heat of VbO₂(l) is 2.0 J/(g · °C). Determine the amount of thermal energy, in J, that must be added to a 5.5 g sample of VbO₂(l) to raise the temperature from 40. °C to 60. °C.
- (c) Two intervals are marked on the graph where the temperature of the sample remains constant. Given that the sample is heated at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid H₂O (boiling point = 100 °C) are weaker than those in liquid VbO₂. Do you agree or disagree with the student's claim? Justify your answer in terms of the relative boiling points.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO_2 at 110°C . The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 80°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 7.0 atm , and the initial pressure of $\text{O}_2(g)$ is 5.5 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 2.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a smaller volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{VbO}_2(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

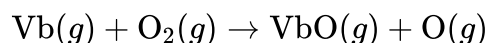
To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	1.2×10^{-3}

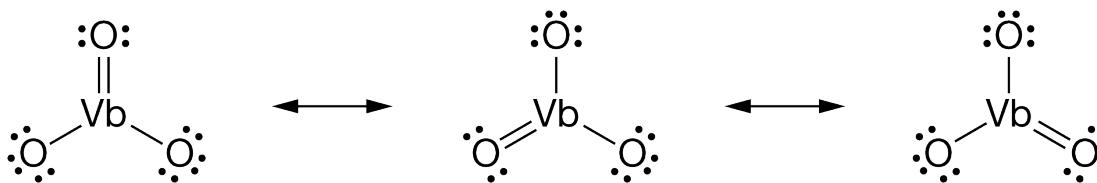
- (i) Based on the data, determine the order of the reaction with respect to $\text{O}_2(g)$. Justify your answer.
- (j) The reaction is known to be first order with respect to $\text{Vb}(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms a molecule with formula VbO_3 . The following Lewis diagrams represent the VbO_3 molecule.



- (l) A student claims that VbO_3 contains all three Vb – O bonds of equal length. Do you agree or disagree? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct hybridization.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

· sp^2

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculation of thermal energy.

· $q = mc\Delta T = (5.5 \text{ g})(2.0 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}})(60. ^\circ\text{C} - 40. ^\circ\text{C}) = 220 \text{ J}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation equivalent to the following.

· The time periods Δt_1 and Δt_2 correspond to the melting and vaporization of VbO_2 , respectively. The data imply that the enthalpy of vaporization of VbO_2 is greater than its enthalpy of fusion. As a result, it requires more energy, and correspondingly a longer period of time, for VbO_2 to completely vaporize than to completely melt.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The higher boiling point of water ($100 ^\circ\text{C}$ vs. $70 ^\circ\text{C}$) indicates that more energy is required for the phase change from liquid to vapor. This difference arises from stronger total intermolecular attractions in water than in VbO_2 .

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides a correct explanation of the error equivalent to the following.

· The maximum of the dashed line should be to the left of the maximum of the solid line (and taller). According to kinetic molecular theory, the average speed of the molecules decreases when the temperature decreases.

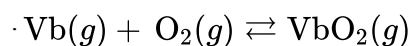
Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct partial pressures.



	Vb(g)	O ₂ (g)	VbO ₂ (g)
Initial Partial Pressure (atm)	7.0	5.5	0.0
Change in Pressure (atm)	-2.5	-2.5	+2.5
Final Partial Pressure (atm)	4.5	3.0	2.5

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The response provides a correct calculation of K_p consistent with the pressures in part (f).

$$K_p = \frac{P_{\text{VbO}_2}}{(P_{\text{Vb}})(P_{\text{O}_2})} = \frac{2.5}{(4.5)(3.0)} = 0.19$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Greater than. When the volume of the system decreases, $Q < K$ and the distribution of species shifts towards the product.

OR

· Greater than. According to Le Châtelier's principle, decreasing the volume of a gaseous equilibrium system will shift the distribution of species towards the side with the fewest number of gas particles. In this case, the shift favors the product.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct order and a valid justification equivalent to the following.

· Zeroth order. Comparing trials 2 and 3, when the initial concentration of O_2 is halved, the initial rate is unchanged.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct rate law consistent with part (i).

$$\text{Rate} = k[\text{Vb}]$$

OR

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$\text{Rate} = k [\text{Vb}] [\text{O}_2]^0$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a justification, consistent with part (j), equivalent to the following.

· No. The elementary reaction involves a bimolecular collision of one Vb and one O₂ particle, which is inconsistent with the zeroth-order dependence on O₂ established in parts (i) and (j).

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a valid justification equivalent to the following.

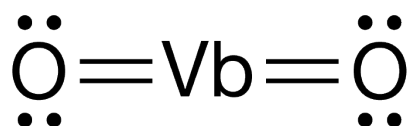
· Agree. The actual arrangement of electrons in VbO₃ is a blend of the three equivalent resonance structures. Thus, all three Vb – O bonds are equal in length.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

26. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

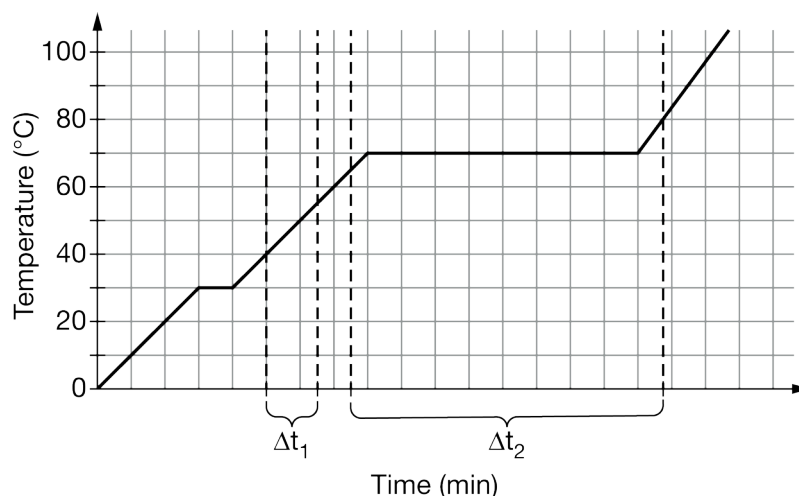
Answer the following questions about the compounds and reactions of Vb, a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VbO₂, an oxide of Vb.



- (a) Identify the hybridization of the valence orbitals of the Vb atom in VbO₂.

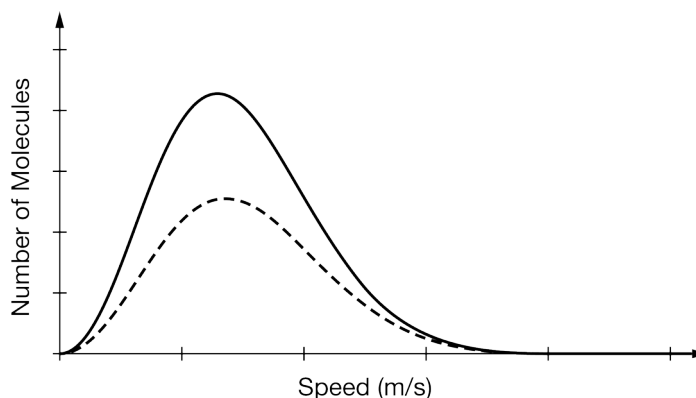
A student investigates the physical properties of VbO₂ and finds the following graph where thermal energy is added at a constant rate to a sample of VbO₂, which is initially a solid at 0 °C. The pressure is held constant at 1.00 atm throughout the heating process.



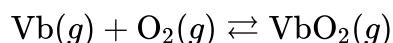
- (b) The student determines that the specific heat of VbO₂ is 2.0 J/(g · °C). Determine the amount of thermal energy, in J, that must be added to a 4.5 g sample of VbO₂(l) to raise the temperature from 40. °C to 55 °C.
- (c) The two intervals marked on the graph correspond to the time required for the temperature of the sample to increase by 15 °C. Given that the sample is heated at a constant rate, explain why $\Delta t_2 > \Delta t_1$.
- (d) The student claims that the intermolecular attractions in liquid H₂O ($\Delta H_{vap}^\circ = 40$ kJ/mol) are stronger than those in liquid VbO₂ ($\Delta H_{vap}^\circ = 32$ kJ/mol). Do you agree or disagree with the student's claim? Justify your answer in terms of the relative values of ΔH_{vap}° .

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VbO_2 at 110°C . The student incorrectly sketches the dashed line to show the distribution of VbO_2 particles when the temperature decreases to 80°C . Explain the student's error.



The student injects $\text{Vb}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vb}(g)$ is 6.0 atm , and the initial pressure of $\text{O}_2(g)$ is 4.0 atm . $\text{Vb}(g)$ and $\text{O}_2(g)$ react to produce $\text{VbO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 2.5 atm of $\text{VbO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vb}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a larger volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{VbO}_2(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

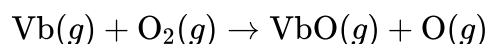
To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vb}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vb}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	8.0×10^{-4}
2	0.20	0.30	1.2×10^{-3}
3	0.10	0.30	6.0×10^{-4}

Chapter 10 Part 2: Le Chatelier's Principle FRQ

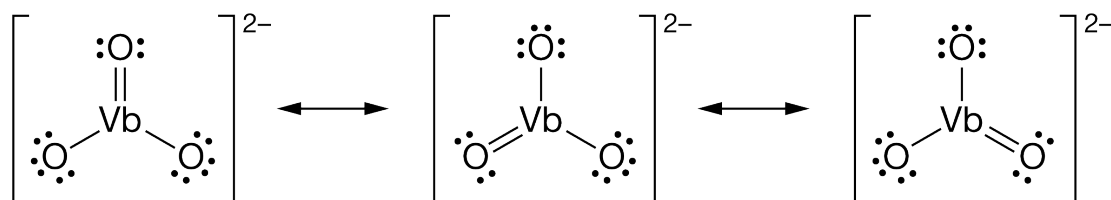
- (i) Based on the data, determine the order of the reaction with respect to $O_2(g)$. Justify your answer.
- (j) The reaction is known to be first order with respect to $Vb(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element Vb forms ions with formula VbO_3^{2-} . The following Lewis diagrams represent the VbO_3^{2-} ion.



- (l) A student claims that VbO_3^{2-} contains two $Vb - O$ bonds of equal length and one $Vb - O$ bond that is shorter than the other two. Do you agree or disagree? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct hybridization.

sp

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculation of thermal energy.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$q = mc\Delta T = (4.5 \text{ g})(2.0 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}})(55^\circ\text{C} - 40^\circ\text{C}) = 135 \text{ J} \approx 140 \text{ J}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation equivalent to the following.

The substance undergoes a phase change (liquid to gas) during Δt_2 but not during Δt_1 . The phase change absorbs energy without changing the temperature of the substance, so the time interval for the sample to increase by 15°C is longer in Δt_2 .

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

Agree. The greater heat of vaporization for water (40 kJ/mol vs. 32 kJ/mol) indicates that more energy is required for the phase change from liquid to vapor. This difference arises from stronger total intermolecular attractions in water than in VbO_2 .

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation of the error equivalent to the following.

The maximum of the dashed line should be to the left of the maximum of the solid line (and taller). According to kinetic molecular theory, the average speed of the molecules decreases when the temperature decreases.

Part F

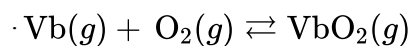
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides the correct partial pressures.



	Vb(g)	O ₂ (g)	VbO ₂ (g)
Initial Partial Pressure (atm)	6.0	4.0	0.0
Change in Pressure (atm)	−2.5	−2.5	+2.5
Final Partial Pressure (atm)	3.5	1.5	2.5

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct calculation of K_p consistent with the pressures in part (f).

$$K_p = \frac{P_{\text{VbO}_2}}{(P_{\text{Vb}})(P_{\text{O}_2})} = \frac{2.5}{(3.5)(1.5)} = 0.48$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Less than. When the volume of the system increases, $Q > K$ and the distribution of species shifts towards reactants.

OR

Chapter 10 Part 2: Le Chatelier's Principle FRQ

· Less than. According to Le Châtelier's principle, increasing the volume of a gaseous equilibrium system will shift the distribution of species towards the side with the greatest number of gas particles. In this case, the shift favors the reactants.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct order and a valid justification equivalent to the following.

· First order. Comparing trials 2 and 3, when the initial concentration of O_2 is halved, the initial rate is also halved.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct rate law consistent with part (i).

· $\text{Rate} = k [\text{Vb}][O_2]$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a justification, consistent with part (j), equivalent to the following.

· Yes. The elementary reaction involves a bimolecular collision of one Vb and one O_2 particle, which corresponds to the respective order of each reactant in the rate law expression.

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0

1

The response provides both a correct answer and a valid justification equivalent to the following.

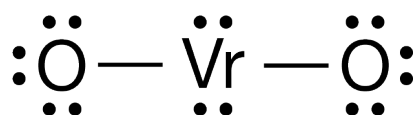
· Disagree. The actual arrangement of electrons in VbO_3^{2-} is a blend of the three equivalent resonance structures. Thus, all three $\text{Vb} - \text{O}$ bonds are equal in length.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

27. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

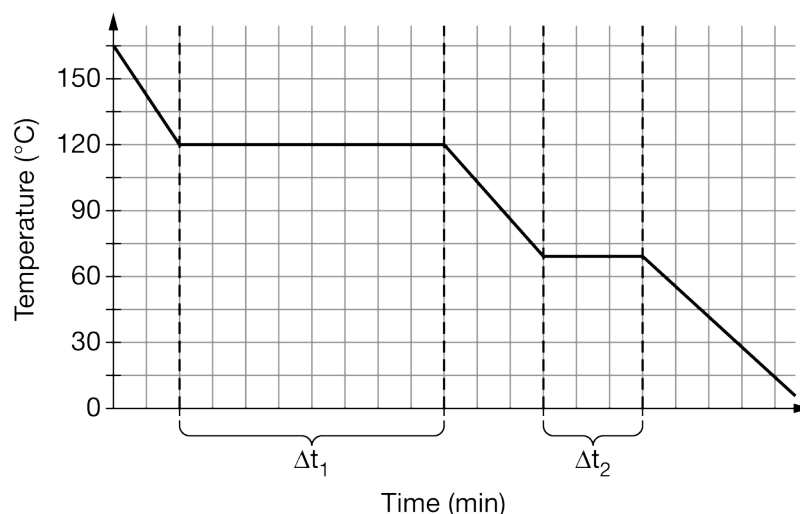
Answer the following questions about the compounds and reactions of Vr , a newly discovered nonmetallic element.

The following is a complete Lewis electron-dot diagram for VrO_2 , an oxide of Vr .



- (a) Based on VSEPR theory, predict the approximate bond angle in VrO_2 .

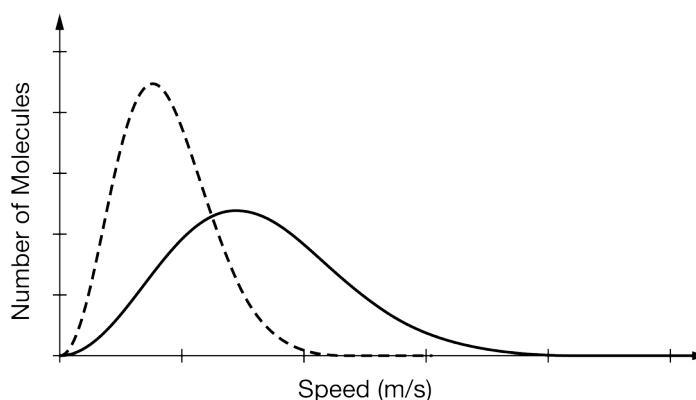
A student investigates the physical properties of VrO_2 and finds the following graph where thermal energy is removed at a constant rate from a sample of VrO_2 , which is initially a gas at 165°C . The pressure is held constant at 1.00 atm throughout the cooling process.



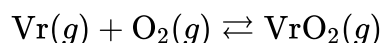
- (b) The student determines that the specific heat of $\text{VrO}_2(l)$ is $2.0\text{ J}/(\text{g} \cdot ^\circ\text{C})$. Determine the amount of thermal energy, in J, that must be removed from a 25 g sample of $\text{VrO}_2(l)$ to reduce the temperature from 115°C to 105°C .
- (c) Two intervals are marked on the graph where the temperature of the sample remains constant. Given that the sample is cooled at a constant rate, explain why $\Delta t_1 > \Delta t_2$.
- (d) The student claims that the intermolecular attractions in liquid SO_2 (boiling point = -10°C) are weaker than those in liquid VrO_2 . Do you agree or disagree with the student's claim? Justify your answer in terms of the relative boiling points.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

(e) The solid curve below shows the distribution of molecular speeds for a sample of gaseous VrO_2 at 165°C . The student incorrectly sketches the dashed line to show the distribution of VrO_2 particles when the temperature increases to 205°C . Explain the student's error.



The student injects $\text{Vr}(g)$ and $\text{O}_2(g)$ into a rigid container at a constant temperature of 400 K . The initial pressure of $\text{Vr}(g)$ is 7.0 atm , and the initial pressure of $\text{O}_2(g)$ is 5.5 atm . $\text{Vr}(g)$ and $\text{O}_2(g)$ react to produce $\text{VrO}_2(g)$, as represented by the following equation.



(f) When the system reaches equilibrium, 1.5 atm of $\text{VrO}_2(g)$ is present in the container. Determine the equilibrium partial pressures of $\text{Vr}(g)$ and $\text{O}_2(g)$ in the container.

(g) What is the value of the equilibrium constant, K_p , for the reaction at 400 K ?

(h) The student transfers the equilibrium mixture to a new container with a smaller volume at 400 K . After equilibrium is re-established, would the mole fraction of $\text{Vr}(g)$ be less than, equal to, or greater than what it was in the original container? Justify your answer.

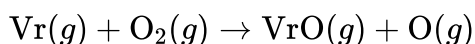
To study the kinetics of the reaction, the student runs several trials of the reaction between $\text{Vr}(g)$ and $\text{O}_2(g)$. The data are presented in the following table.

Trial	Initial $[\text{O}_2(g)]$ (M)	Initial $[\text{Vr}(g)]$ (M)	Initial Rate (M/s)
1	0.20	0.20	1.2×10^{-3}
2	0.20	0.40	1.2×10^{-3}
3	0.10	0.40	6.0×10^{-4}

Chapter 10 Part 2: Le Chatelier's Principle FRQ

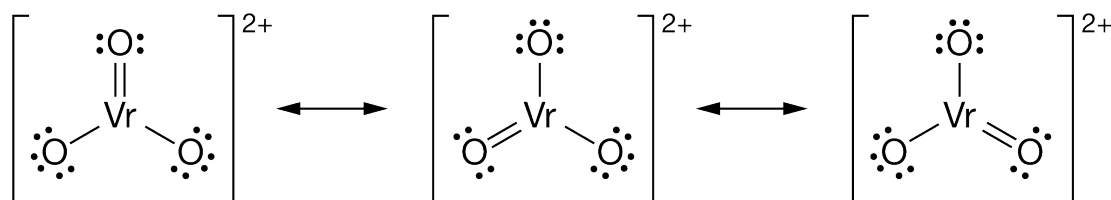
- (i) Based on the data, determine the order of the reaction with respect to $\text{Vr}(g)$. Justify your answer.
- (j) The reaction was found to be first order with respect to $\text{O}_2(g)$. Using this information and the answer to part (i), write the rate law for the reaction.

The student proposes that the following elementary reaction is the rate-limiting step in a multistep mechanism.



- (k) Is the proposed rate-limiting step consistent with the rate law written in part (j)? Justify your answer.

Element **Vr** forms ions with formula VrO_3^{2+} . The following Lewis diagrams represent the VrO_3^{2+} ion.



- (l) A student claims that VrO_3^{2+} contains three $\text{Vr} - \text{O}$ bonds of equal strength. Do you agree or disagree? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct bond angle.

- Any value between 104° and 110°

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculation of thermal energy.

Chapter 10 Part 2: Le Chatelier's Principle FRQ

$$q = mc\Delta T = (25 \text{ g})(2.0 \frac{\text{J}}{\text{g}\cdot^\circ\text{C}})(105^\circ\text{C} - 115^\circ\text{C}) = -500 \text{ J, so 500 J is removed.}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation equivalent to the following.

The time periods Δt_1 and Δt_2 correspond to the condensation and freezing of VrO_2 , respectively. The data imply that the enthalpy of vaporization of VrO_2 is greater than its enthalpy of fusion. As a result, more energy needs to be removed from the system, and correspondingly it takes a longer period of time, for VrO_2 to completely condense than to completely freeze.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

Agree. The lower boiling point of SO_2 (-10°C vs. 120°C) indicates that less energy is required for the phase change from liquid to vapor. This difference arises from weaker total intermolecular attractions in SO_2 than in VrO_2 .

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct explanation of the error equivalent to the following.

The maximum of the dashed line should be to the right of the maximum of the solid line. According to kinetic molecular theory, the average speed of the molecules increases when the temperature increases.

Part F

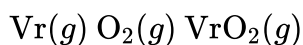
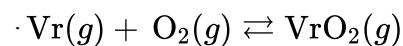
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 10 Part 2: Le Chatelier's Principle FRQ



0	1
---	---

The response provides the correct partial pressures.



Initial Partial Pressure (atm) 7.0 5.5 0.0

Change in Pressure (atm) -1.5 -1.5 $+1.5$

Final Partial Pressure (atm) 5.5 4.0 1.5

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct calculation of K_p consistent with the pressures in part (f).

$$K_p = \frac{P_{\text{VrO}_2}}{(P_{\text{Vr}})(P_{\text{O}_2})} = \frac{1.5}{(5.5)(4.0)} = 0.068$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Less than. When the volume of the system decreases, $Q < K$ and the distribution of species shifts towards the product.

OR

· Less than. According to Le Châtelier's principle, decreasing the volume of a gaseous equilibrium system will shift the distribution of species towards the side with the fewest number of gas particles. In this case, the shift favors the product.

Part I

Chapter 10 Part 2: Le Chatelier's Principle FRQ

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct order and a valid justification equivalent to one of the following.

- Zeroth order. Comparing trials 1 and 2, when the initial concentration of V_r is doubled, the initial rate is unchanged.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct rate law consistent with part (i).

- $\text{Rate} = k [\text{O}_2]$

OR

- $\text{Rate} = k [V_r]^0 [\text{O}_2]$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both a correct answer and a justification, consistent with part (j), equivalent to the following.

- No. The elementary reaction involves a bimolecular collision of one V_r and one O_2 particle, which is inconsistent with the zeroth-order dependence on V_r established in parts (i) and (j).

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

Chapter 10 Part 2: Le Chatelier's Principle FRQ

The response provides both a correct answer and a valid justification equivalent to the following.

· Agree. The actual arrangement of electrons in VrO_3^{2+} is a blend of the three equivalent resonance structures. Thus, all three $\text{Vr} - \text{O}$ bonds are equal in strength.